

E2 - Getting started with the development ecosystem for PSOC™ Edge E84

Customer training workshop (CTW)

2026/09/27

Public

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PSOC™ Edge E84 resources

- PSOC™ Edge E84 MCU Product page and PSOC™ Developer Journey on Infineon.com
- Datasheets
- Architecture and registers reference manuals
- Application notes
- Code examples
- Development kits
- KBAs and community forum

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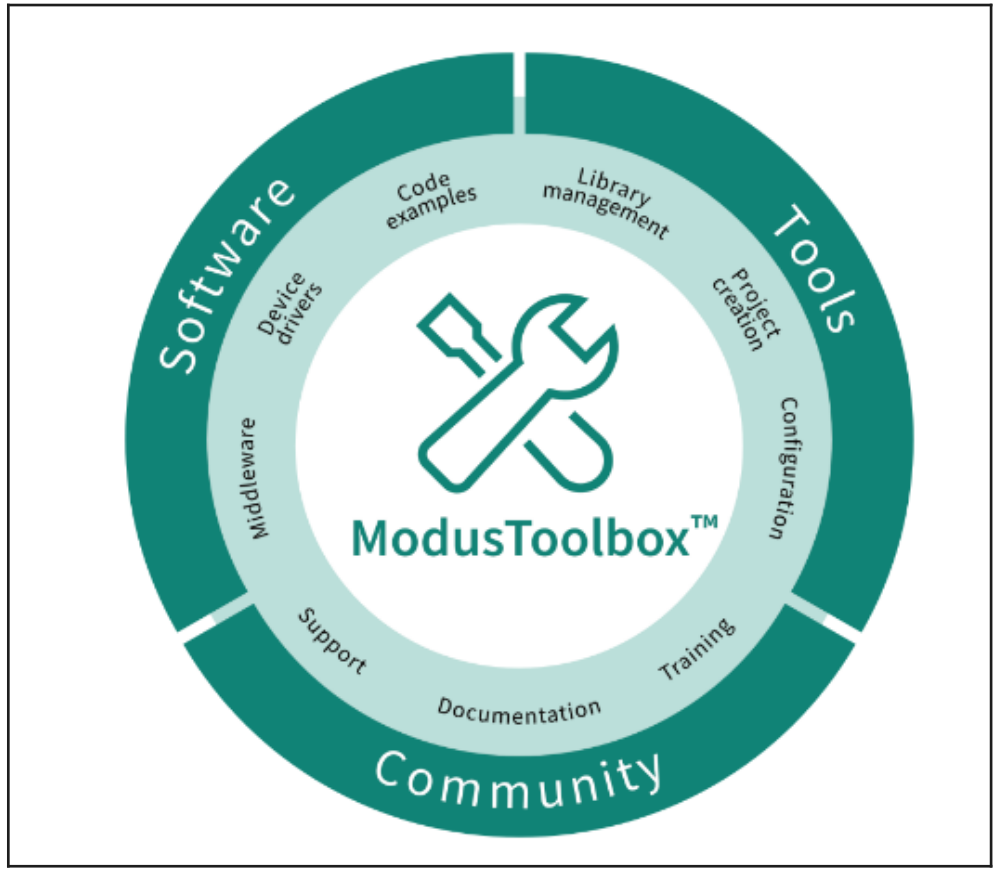
Firmware/application development

ModusToolbox™ provides

- Development environment for Infineon MCUs
- Flexible set of tools for project and system-level configuration
- Low-level drivers
- Libraries
- Operating system support Linux, macOS, and Windows

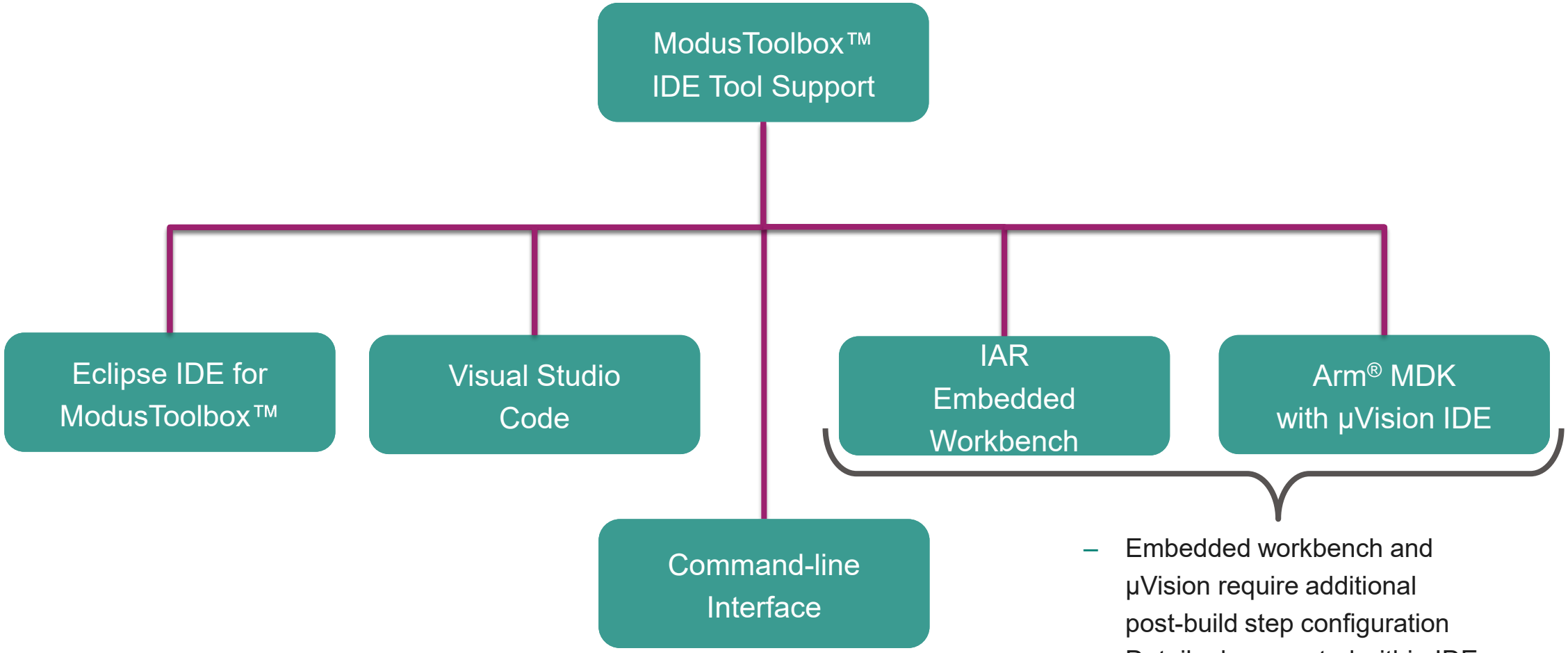
For more details, go to:

https://www.infineon.com/ModusToolboxInstallguide



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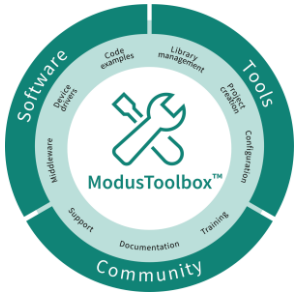
Support for IDEs



- Embedded workbench and µVision require additional post-build step configuration
- Details documented within IDE respective user manuals

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Understanding ModusToolbox™ and Zephyr



- Provides a targeted ecosystem focused on development of Infineon devices

- IDE support, including Eclipse IDE, VS Code, and third-party IDEs
- Support for GCC, IAR, and Arm compilers using ‘make’ based ModusToolbox™ build system
- Graphical UI configurations tool
- Leverages Infineon-developed low-level drivers, middleware abstraction, middleware, and code examples
- Enables key partner middleware via ModusToolbox™ and Friends program



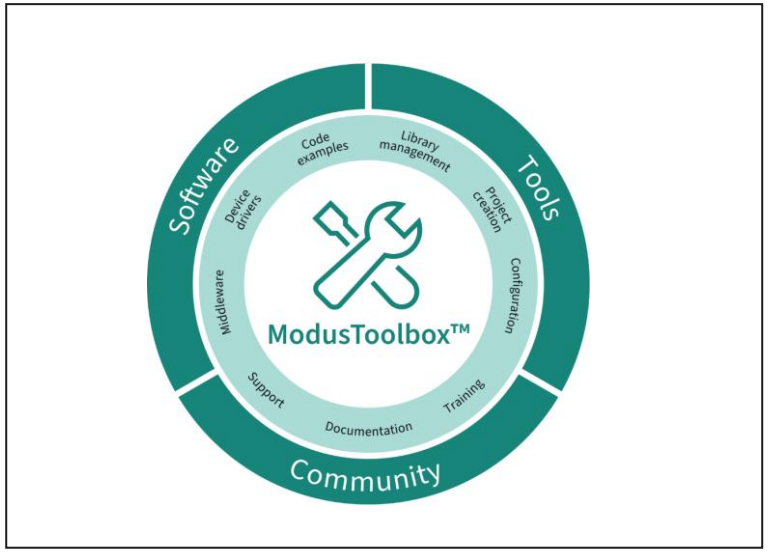
- Provides a community driven ecosystem with support for a targeted set of MCUs from a wide range of silicon providers

- Targeted at command-line users with some community support for VS Code
- Default support for GCC compiler using ‘west’ based Zephyr build environment. Additional toolchains are now being supported
- Text-based configuration via devicetree and Kconfig
- Leverages vendor provided device support; Infineon support based on low-level drivers
- Community contributed middleware, with support for Zephyr defined driver API, stacks, and RTOS

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Overview of DEEPCRAFT™ Studio and ModusToolbox™ workflow

- Installation of ModusToolbox™ and DEEPCRAFT™ supported within the ModusToolbox™ Setup program
- Code Examples aligned with DEEPCRAFT™ Studio Accelerator projects for IMU, Vibration, and Microphone
- Code Examples aligned with DEEPCRAFT™ Studio Data Collection with Graph UX
- Code Examples aligned with DEEPCRAFT™ Model Converter Profiling
- Project connector files in ModusToolbox™ applications with Quick Panel shortcuts for linking to DEEPCRAFT™ Studio Projects



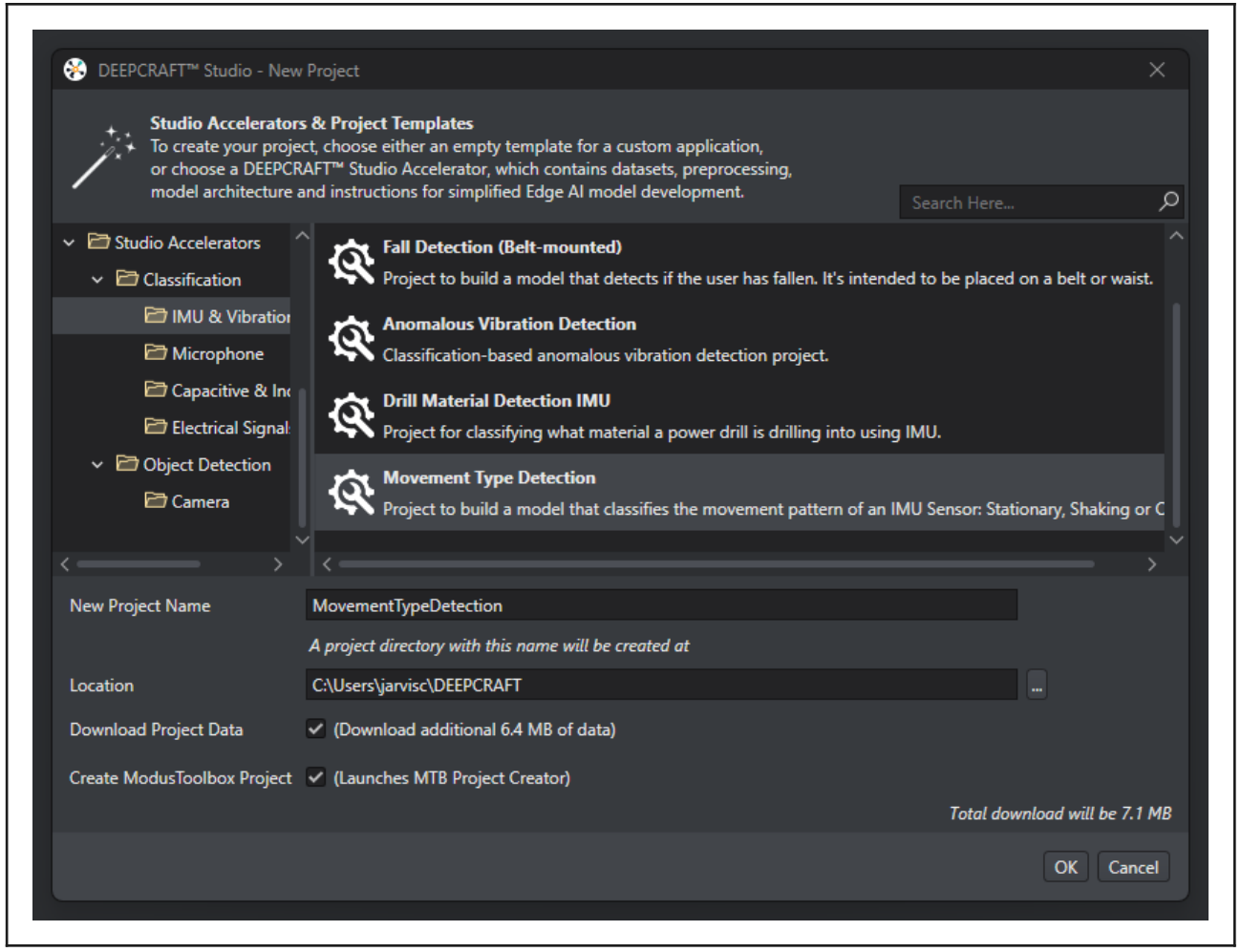
- Integration of ModusToolbox™ Project Creator, supporting coordinated DEEPCRAFT™ Studio and ModusToolbox™ project creation
- Integration ModusToolbox™ Machine Learning Core Tools (ml-coretools), leveraged during Infineon device-specific code generations and optimisation
- ModusToolbox™ project detection and selection during model code generation



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Code Examples with DEEPCRAFT™ and ModusToolbox™

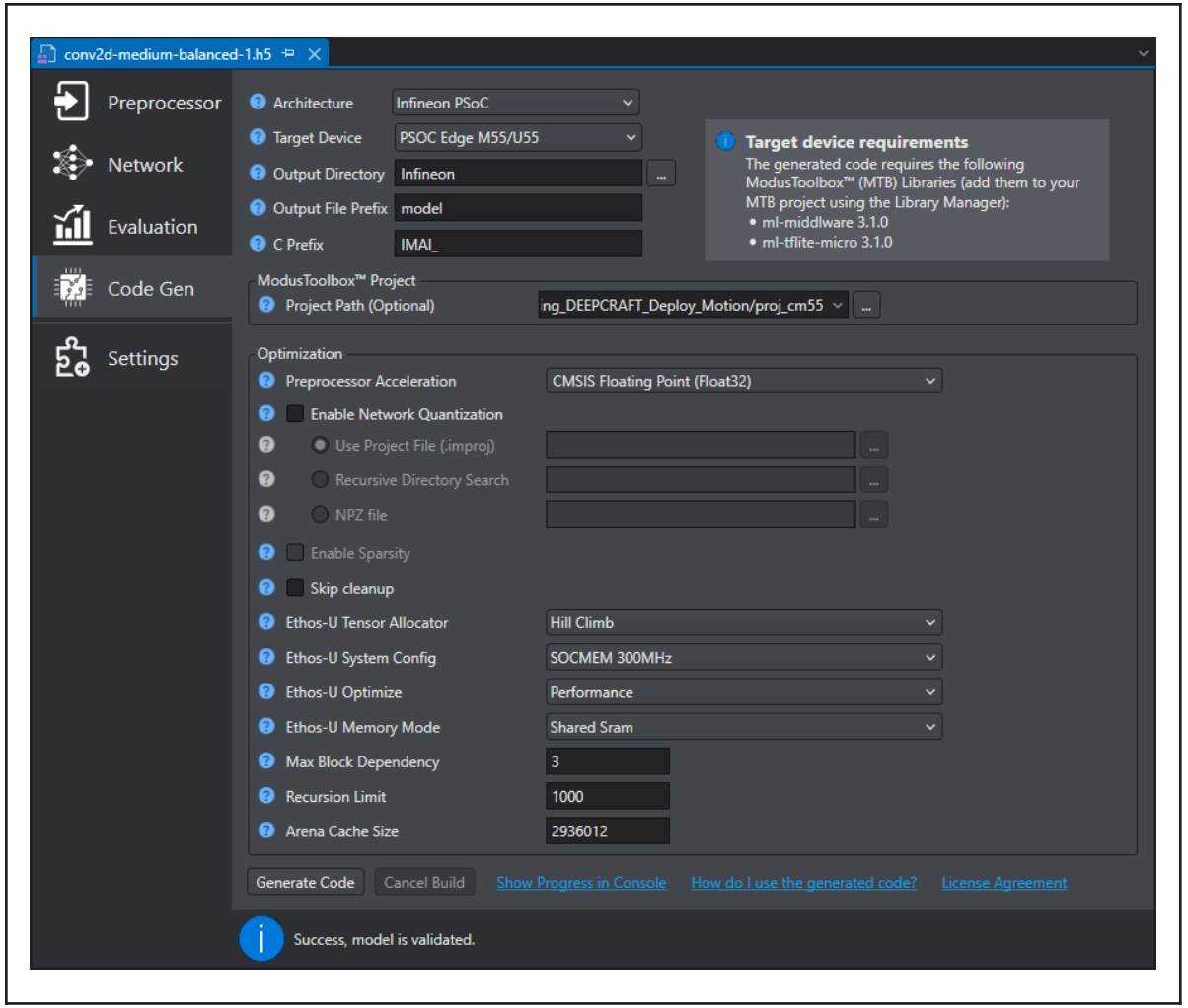
- When creating a new DEEPCRAFT™ Studio project, a ModusToolbox™ project can be created at the same time
- DEEPCRAFT™ Studio includes Studio Accelerator projects that align with ModusToolbox™ code examples, enabling models generated from Studio to be more easily leveraged within a ModusToolbox™ application
- ModusToolbox™ code examples include DEEPCRAFT™ Deploy Audio and Deploy Motions applications



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DEEPCRAFT™ Studio model code generation

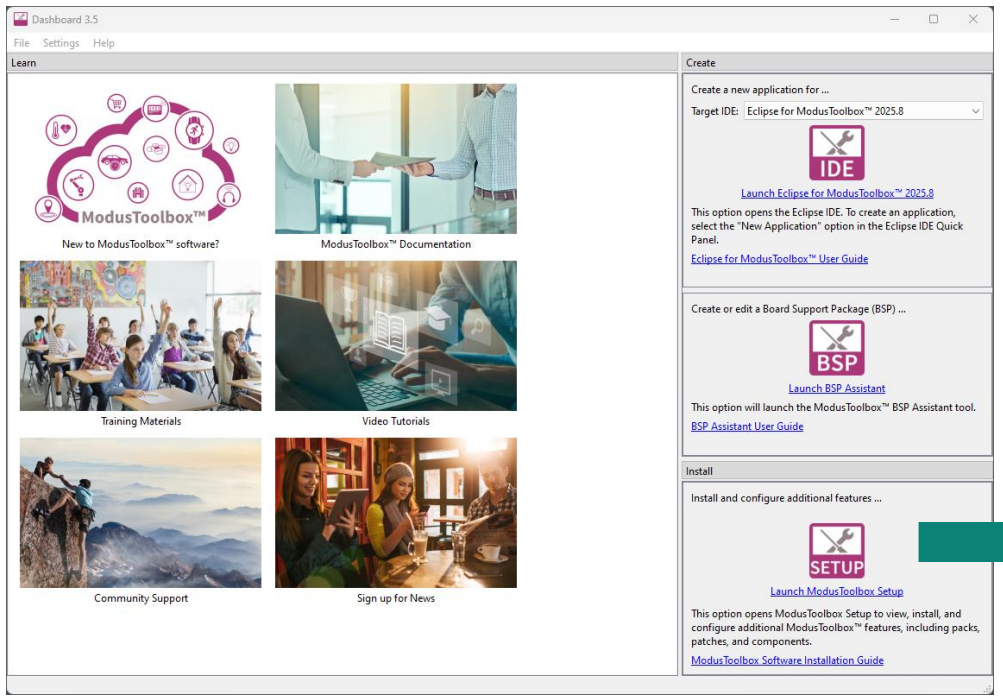
- DEEPCRAFT™ Studio code generation dialog includes pre-defined architecture selections for Infineon devices, including PSOC™ Edge E84
- DEEPCRAFT™ Studio code generation integrates the ModusToolbox™ Machine Learning Core Tools (ml-coretool) with integrated Vela compiler supporting PSOC™ Edge E84
- Selection of Infineon device enables project selection of the connected ModusToolbox™ project and core-specific project path
- Selected ModusToolbox™ project path will be used during the code generation step to enable iterative model development and updates

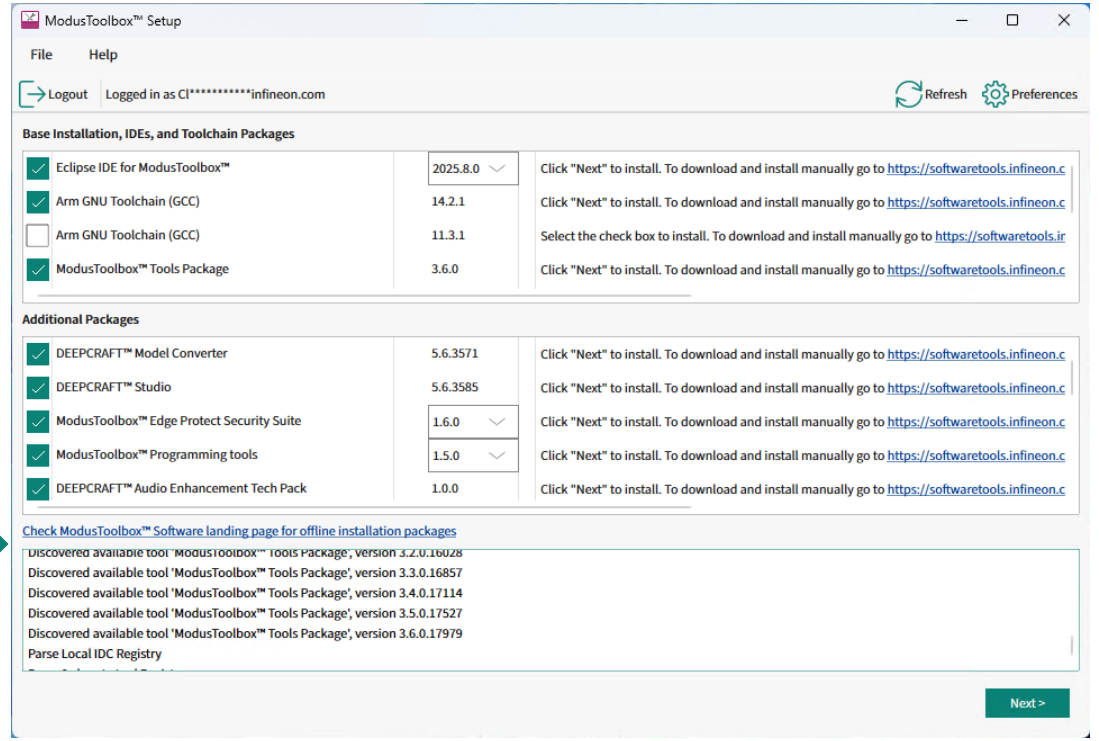


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Installation support within ModusToolbox™ setup

– ModusToolbox™ setup program can be used to install and manage DEEPCRAFT™ applications, including Studio, Model Converter, and Audio Enhancement Tech Pack





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Tools utilized in this presentation

- ModusToolbox™
- Visual Studio Code with the Infineon ModusToolbox™ for VS Code extension
- Edge Protect Security Suite
- ModusToolbox™ Programming Tools
- KIT_PSE84_EVAL_EPC2 BSP
- Terminal Emulator
- KIT_PSE84_EVAL

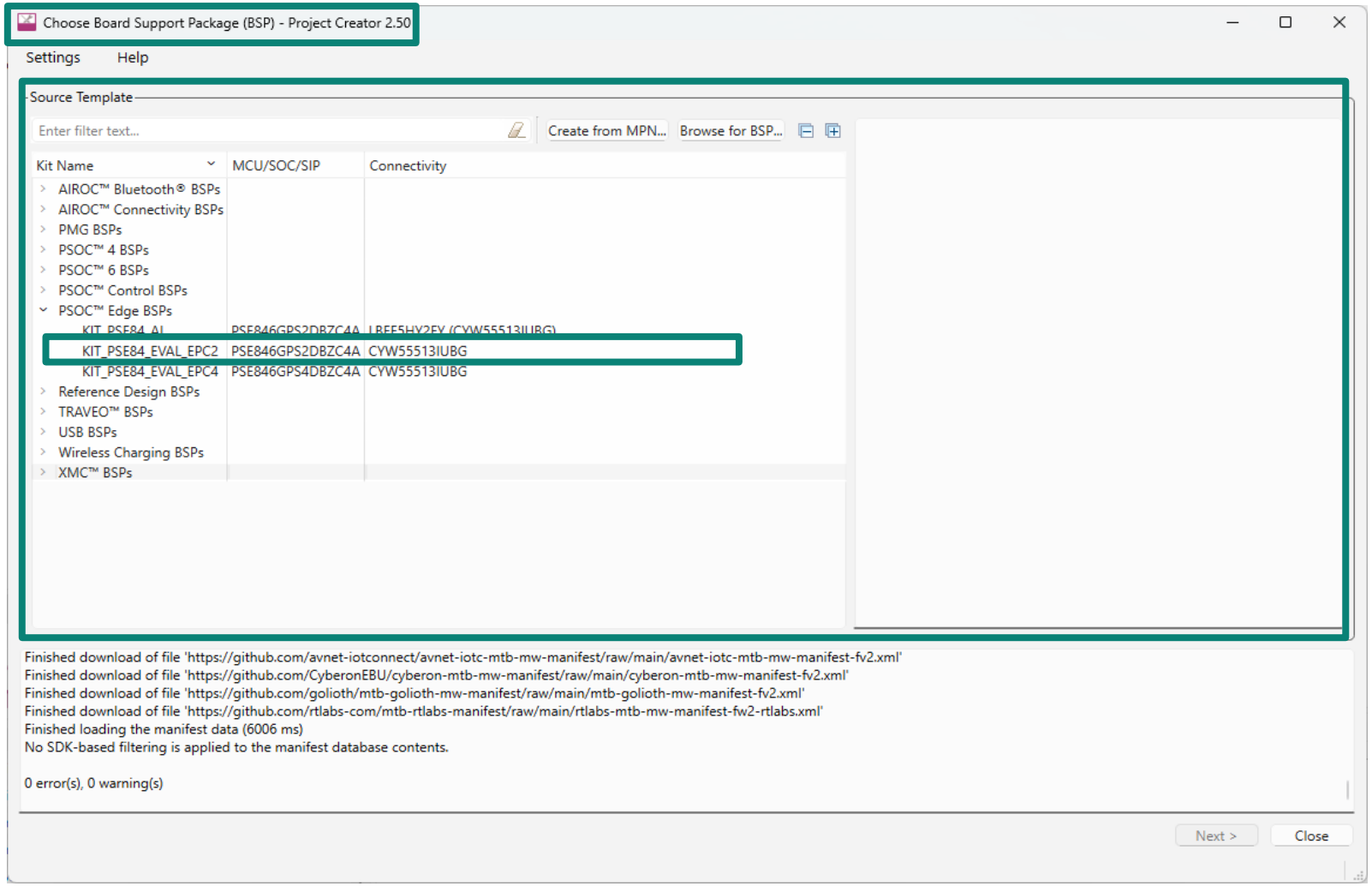
Note: See lab manual for tool installation instructions.

Key documentation leveraged in this video

- **AN235935** – Getting started with PSOC™ Edge E84 on ModusToolbox™ software
- ModusToolbox™ PSE8xxGP Device Support Library (DSL)

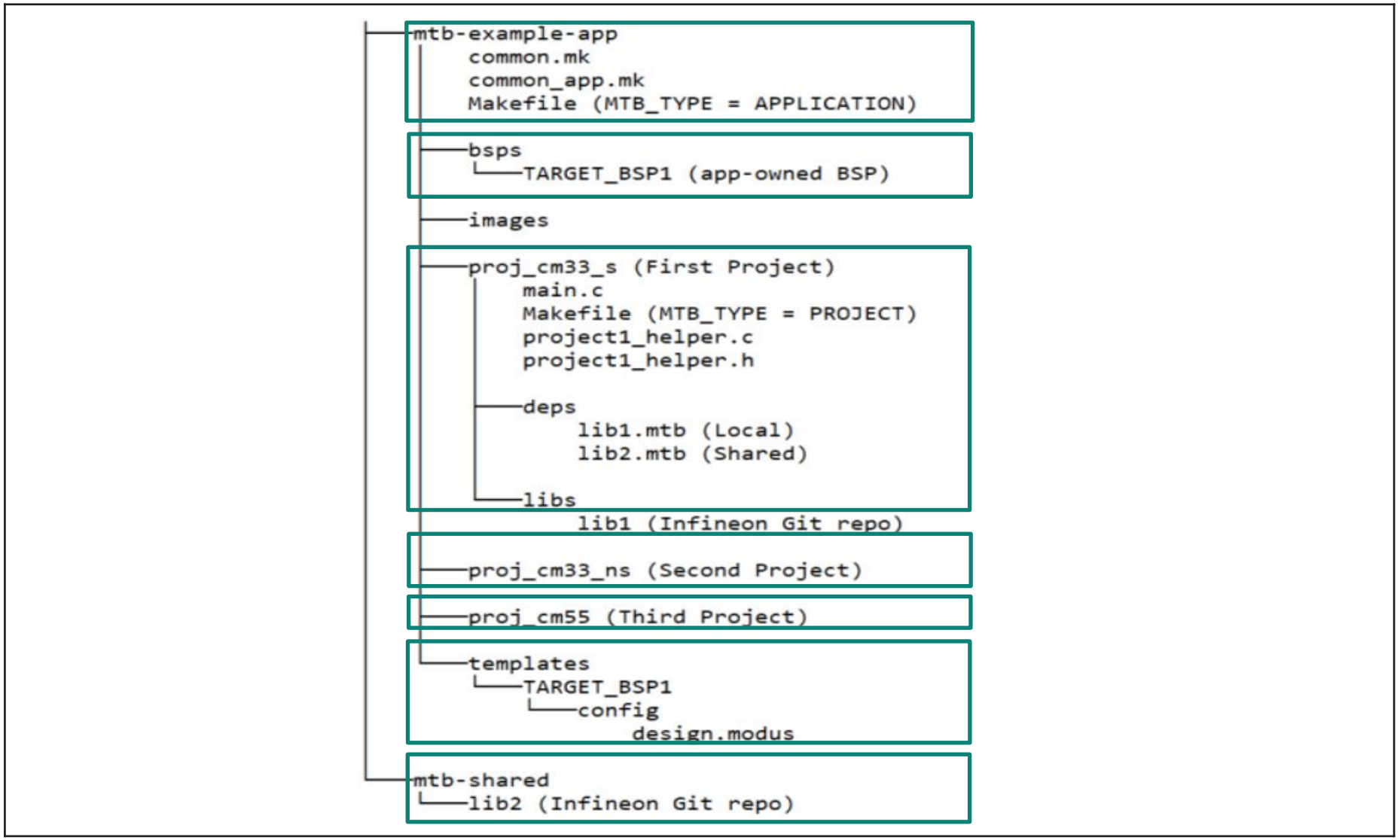
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Project Creator



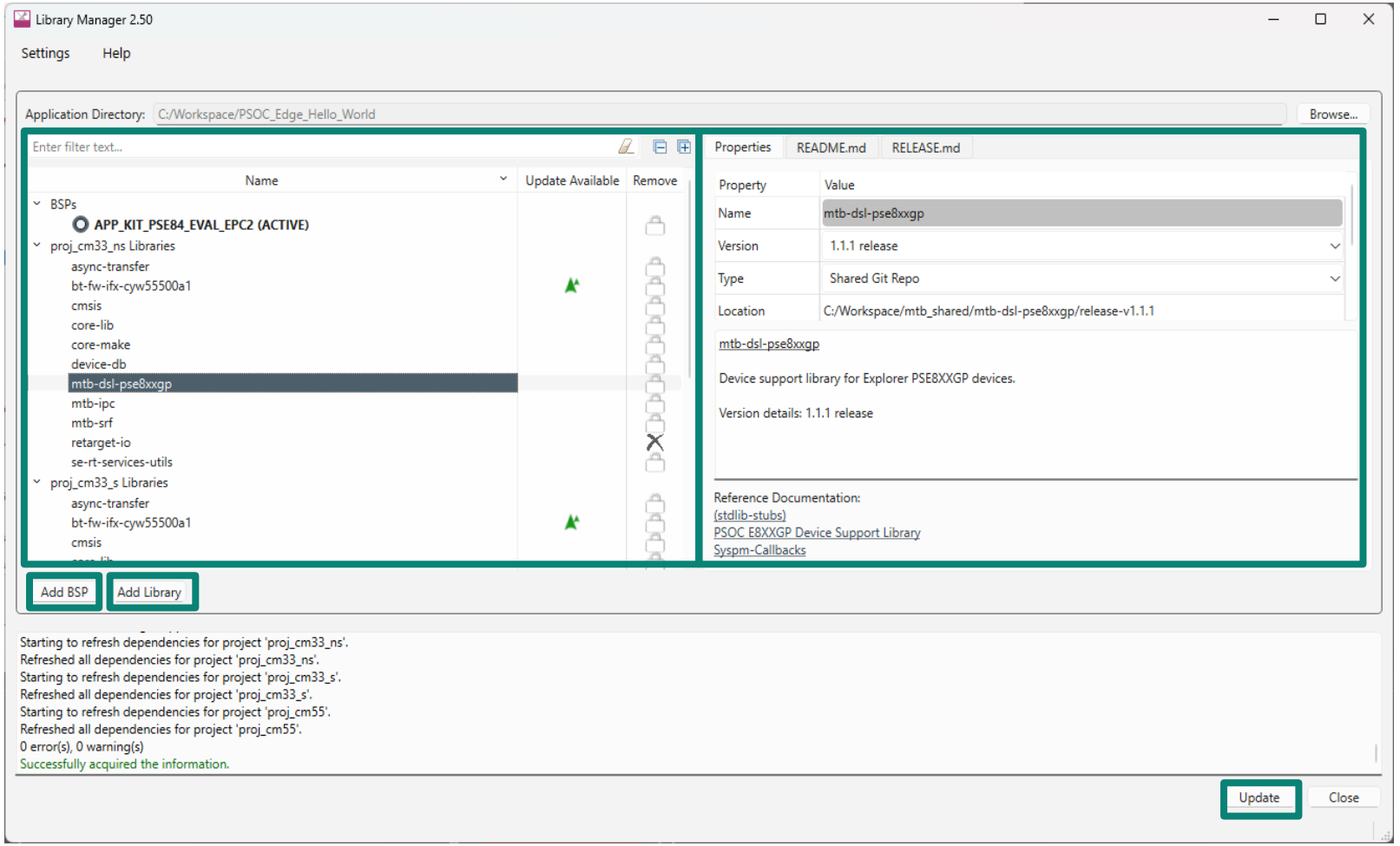
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Understanding the application structure



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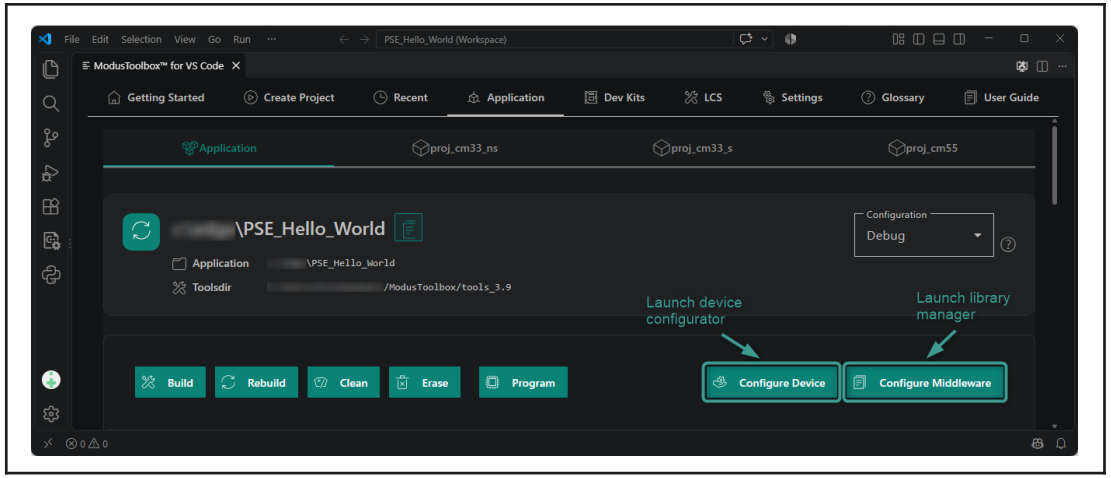
Library Manager

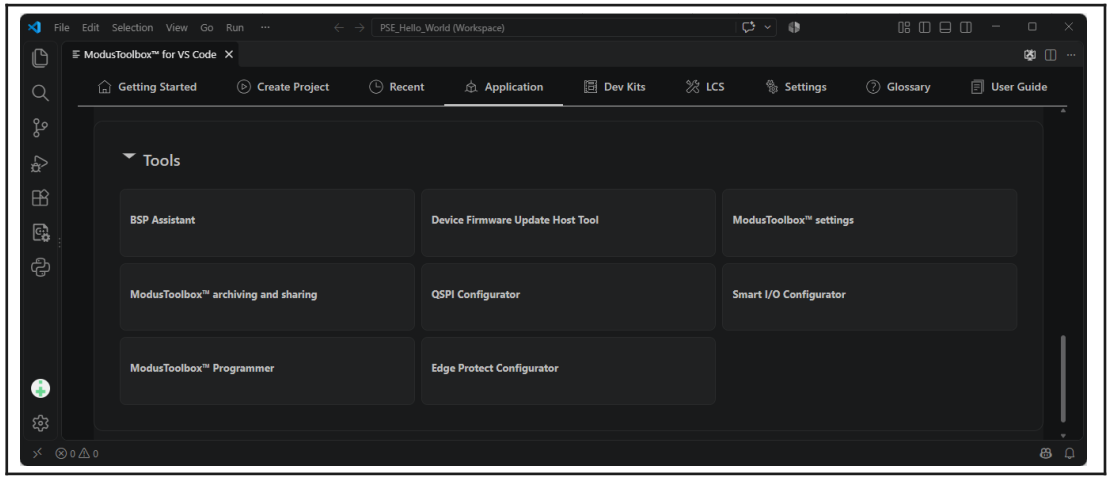


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BSP configurators

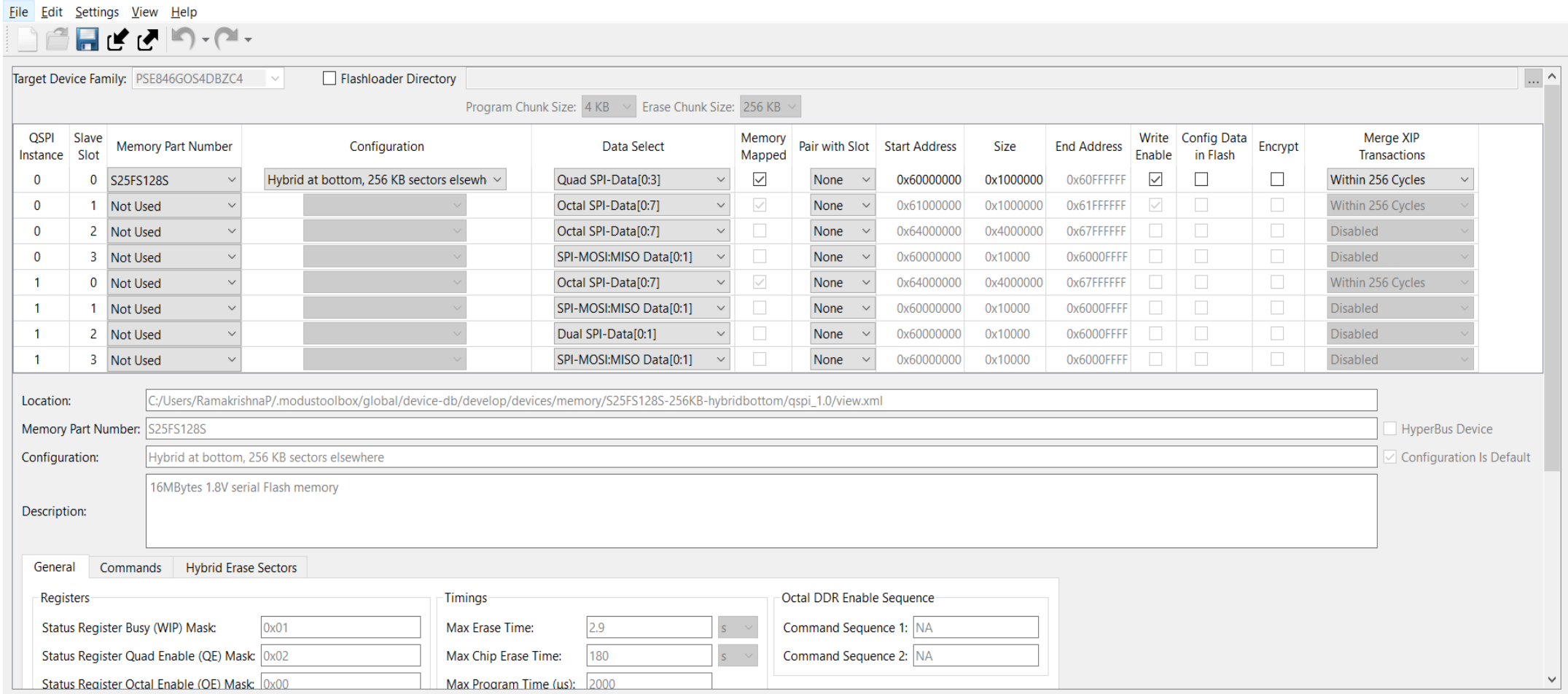
- Device Configurator
- QSPI configurator
- Smart I/O configurator
- Edge protect configurator
- Configurators
  - Creates structures needed for peripherals used by PDL





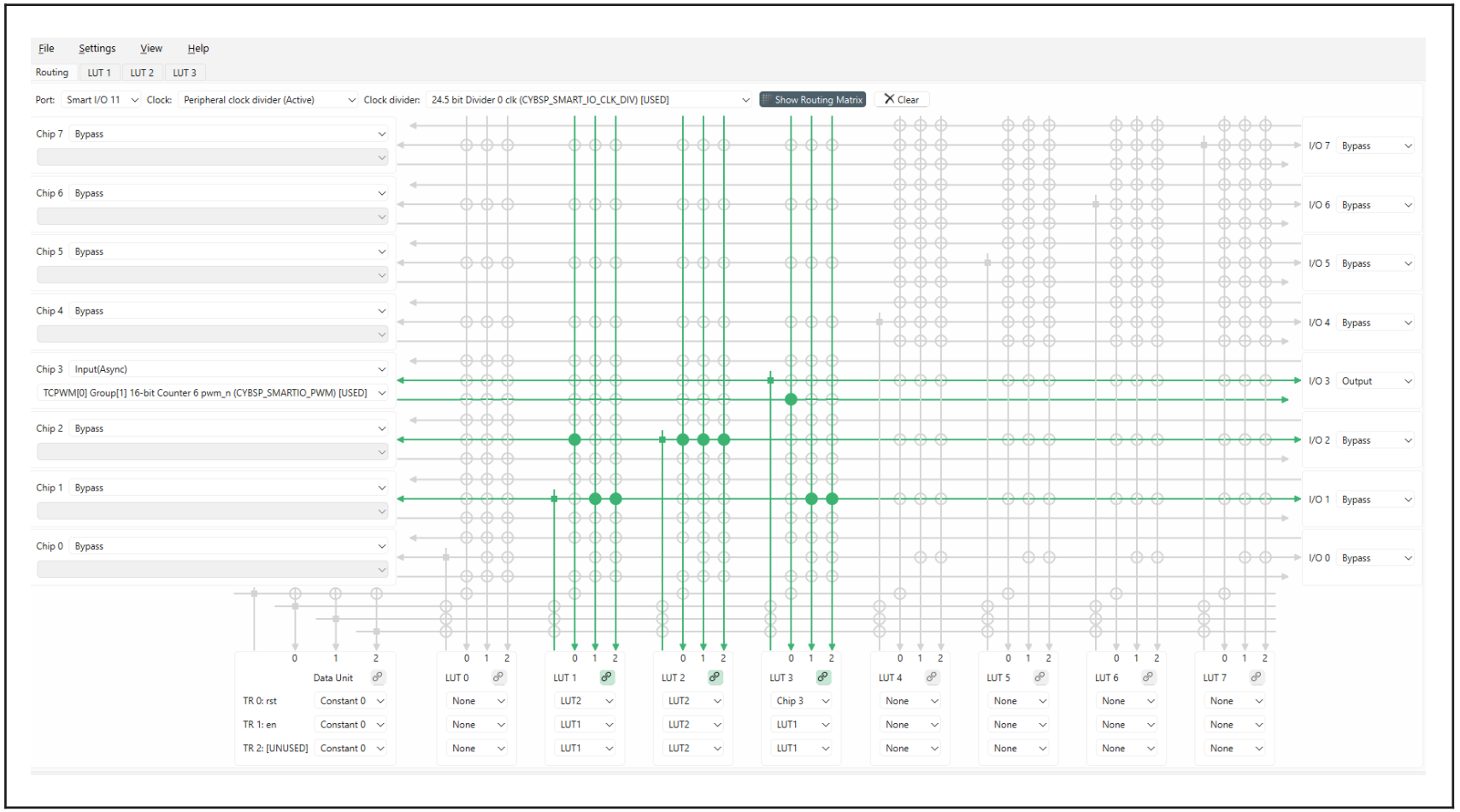
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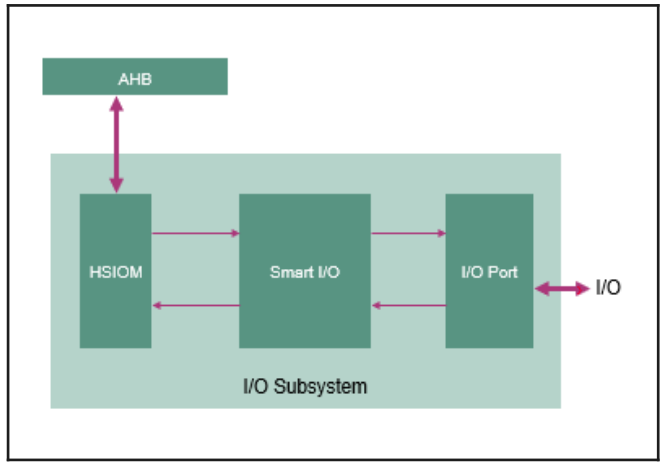
QSPI configurator



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SMART I/O configurator

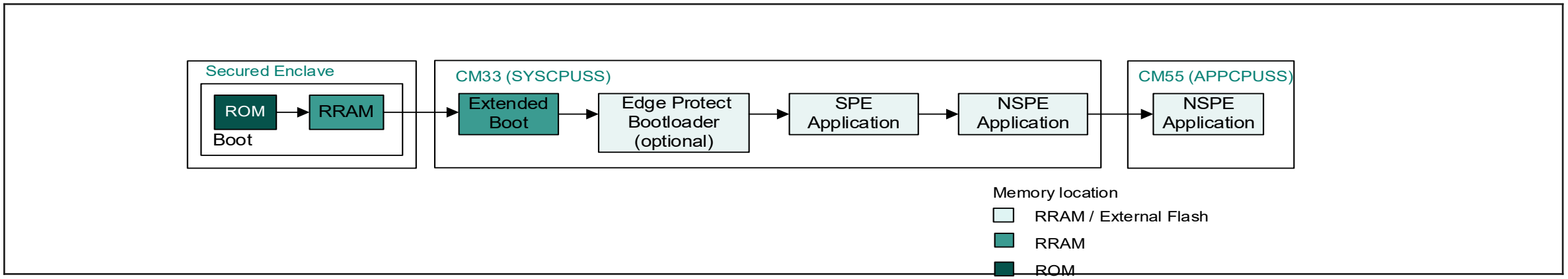




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Application booting sequence

- Boot starts from Secured Enclave as the Root of Trust
- Off-the-shelf extended boot software
  - Boot from RRAM or external flash
  - Alternate serial interface for recovery boot
  - SRAM loading for enhanced security
- Optional MCUboot based Edge Protect Bootloader
- One-time revocable OEM RoT Key



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Device lifecycle and provisioning

Development

– Shipped from Infineon

– Provision with OEM certificate and policy

– For OEM development; until production

Production

– One-time provisioning with “Production” certificate

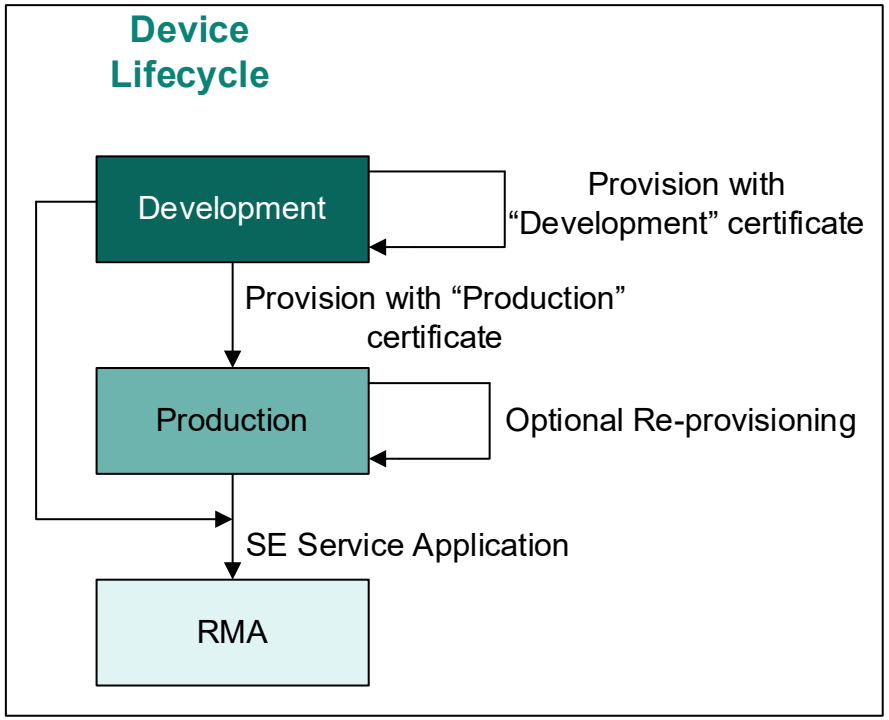
– Re-provisioning if allowed by policy

RMA

– Transition from Development or Production

– Performed by Infineon signed SE Service Application

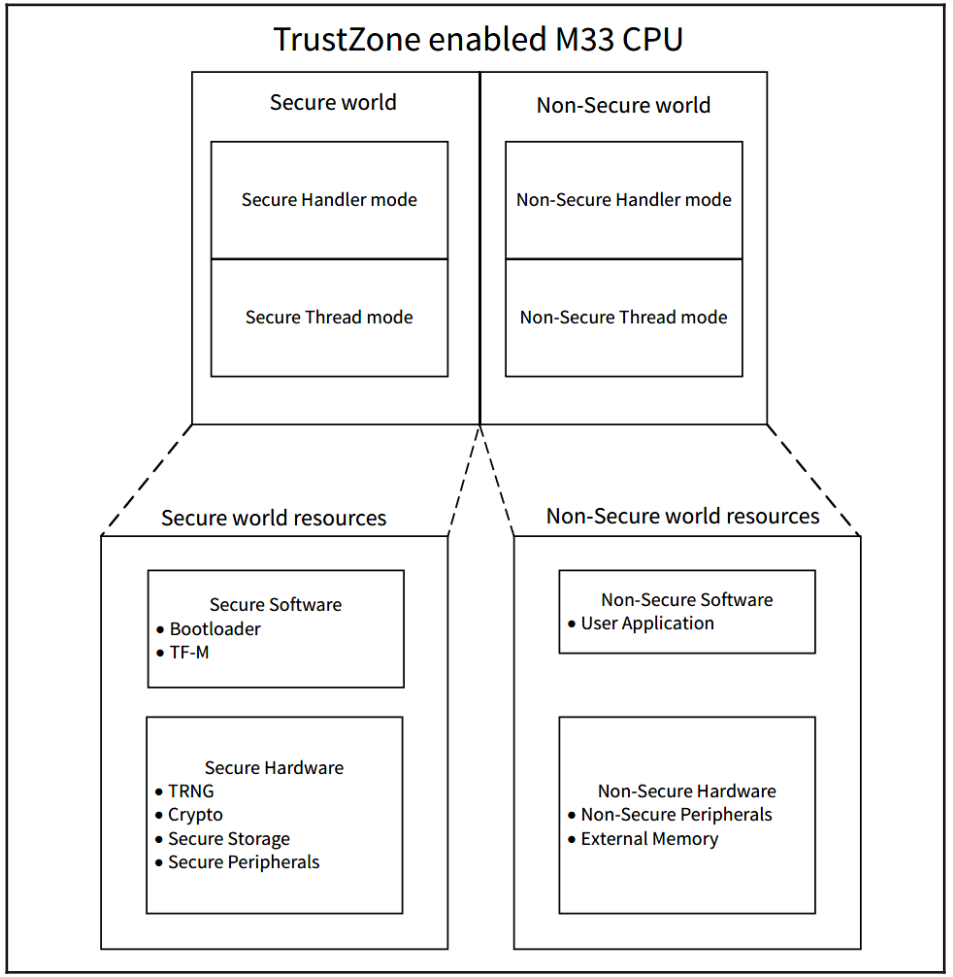
– Devices in RMA mode can be shipped to Infineon for field failure analysis



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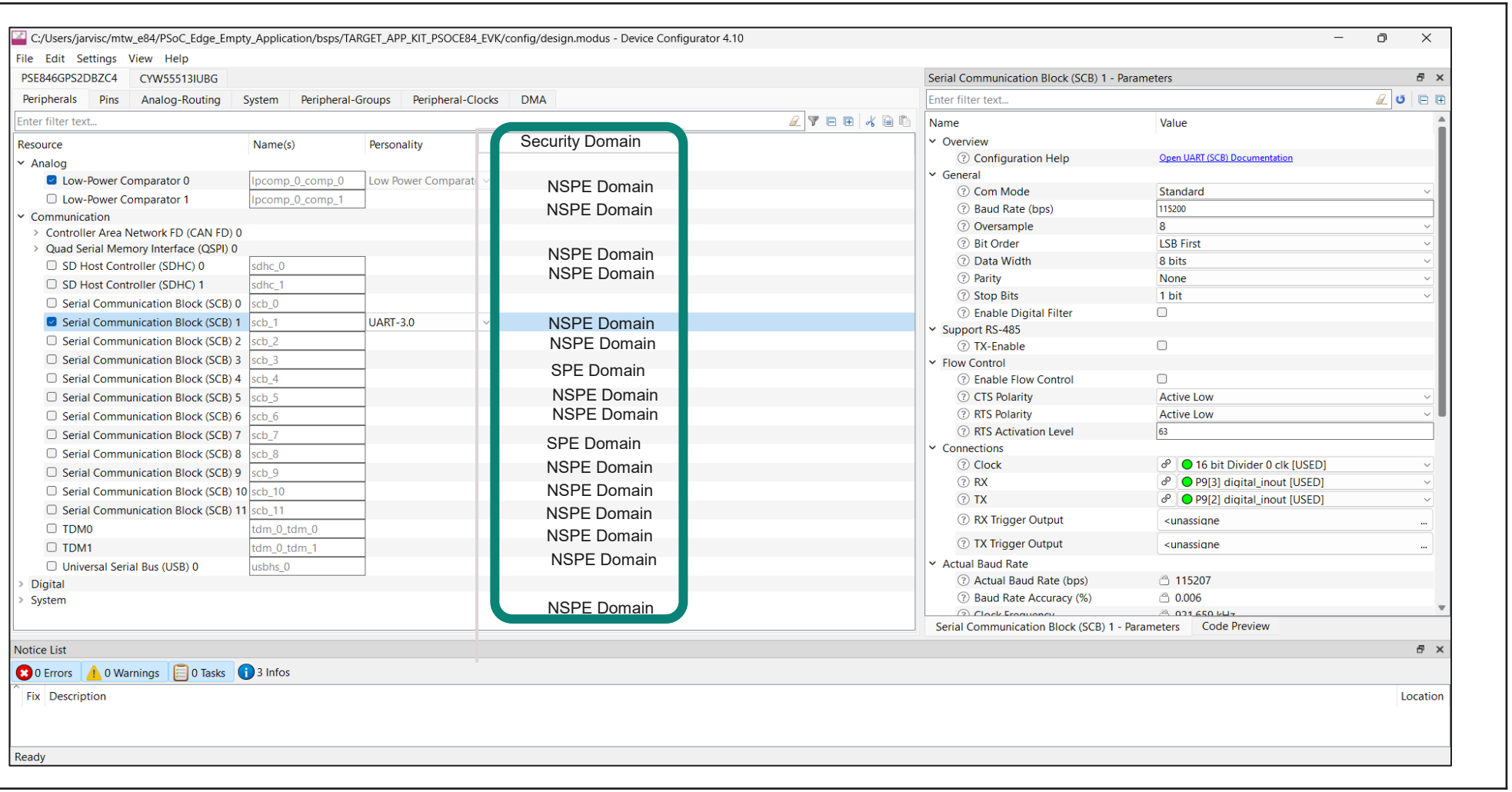
Secure and non-secure operating modes

- Arm® TrustZone - enabled Cortex®-M33 CPU
- Secure and non-secure CPU states
- Attribution units - Secure attribution unit (SAU) and Implementation defined attribution unit (IDAU)
- Banked resources between secure and non-secure states
- Secure and non-secure
  - aliasing of the memory map
  - interrupts
  - peripherals
  - debug



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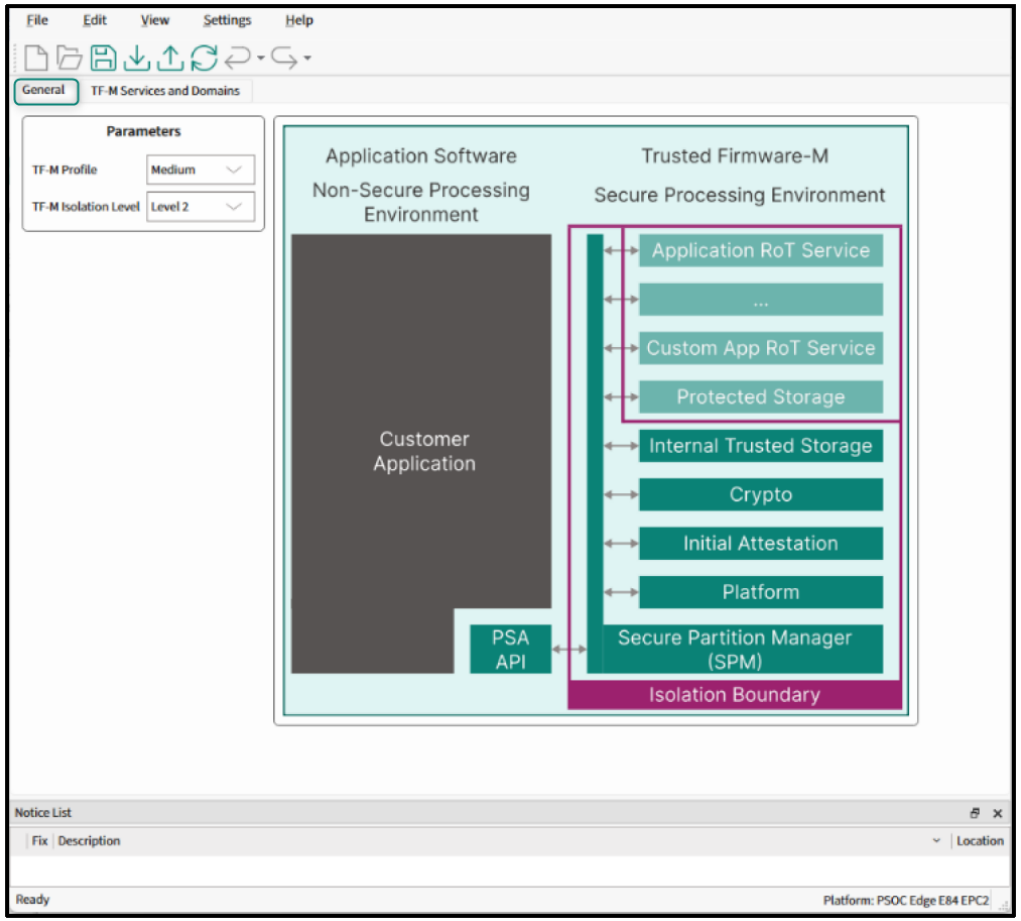
Security domain assignment



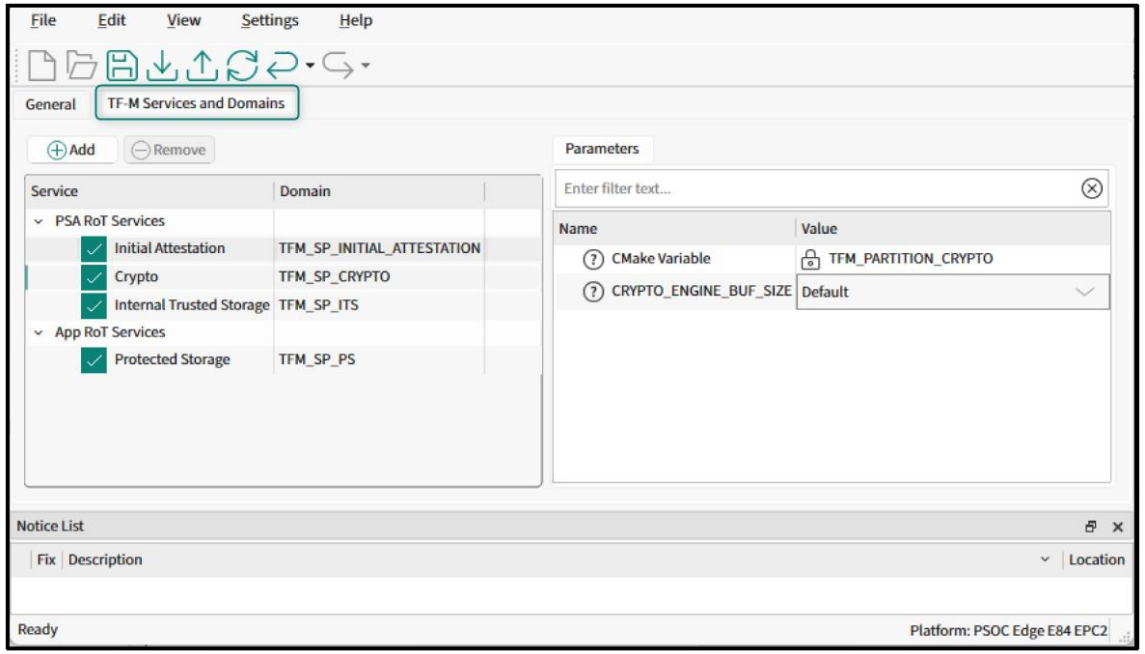
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Edge protect configurator

General tab



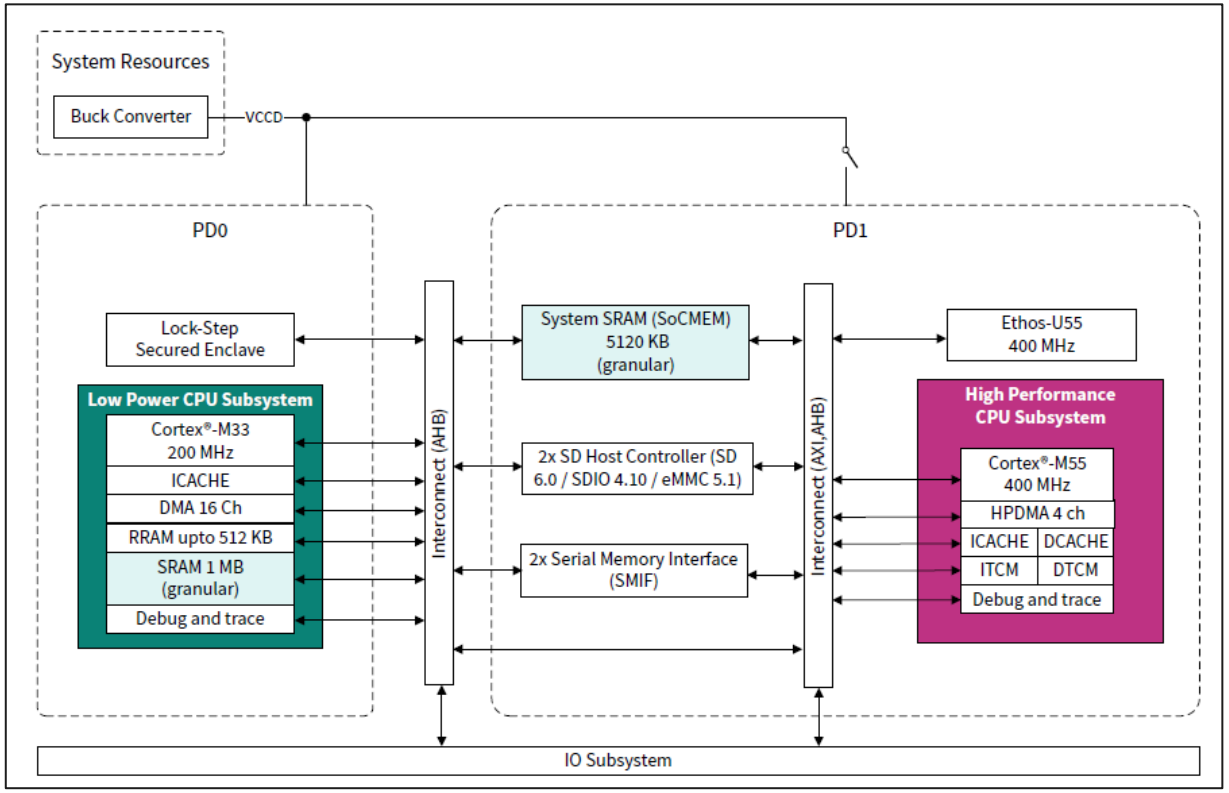
TF-M services and domains tab



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PSOC™ Edge memory architecture

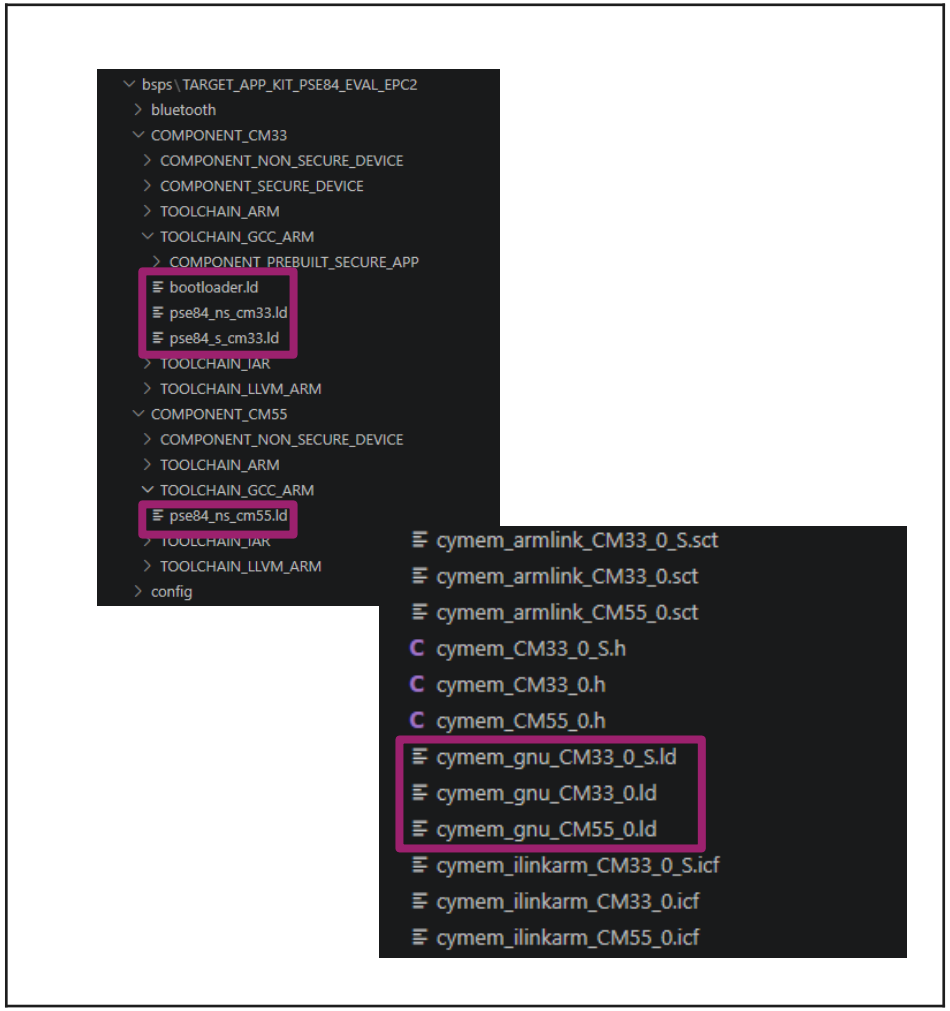
- The PSOC™ Edge features two power domains:
  - PD0: Always-on low-power domain
  - PD1: High-performance domain
- Power source and operating voltage:
  - Both PD0 and PD1 are powered by the internal buck output and operate at same voltage level
- Independent control and flexibility:
  - PD1 can be independently switched off without affecting PD0
  - PD1 can be reactivated as necessary for performance needs
- Power conservation:
  - When high-performance peripherals are not needed, PD1 can be turned off to save power



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Linker scripts

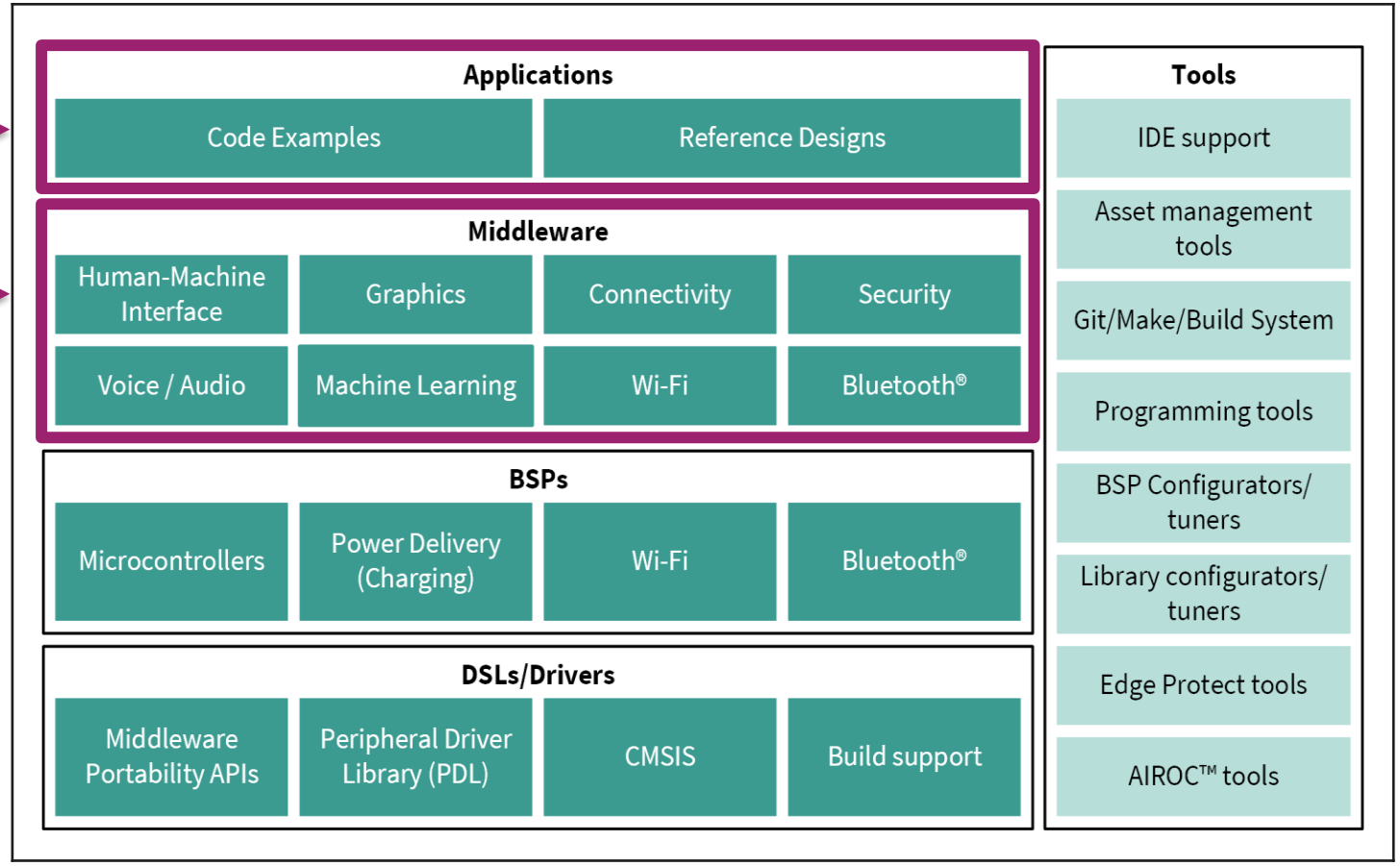
- Four top-level linker files
  - Bootloader
  - CM33 Secure
  - CM33 Non-secure
  - CM55
- Three generated memory *include* files
  - CM33 Secure
  - CM33 Non-secure
  - CM55
- Generated memory *include* files come from the Device Configurator
- New user defined memory regions require direct modification to top-level linker files to leverage generated content



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Firmware/application development :Software resources (1/4)

- Applications
  - Code examples
  - Reference Designs
- Middleware Libraries

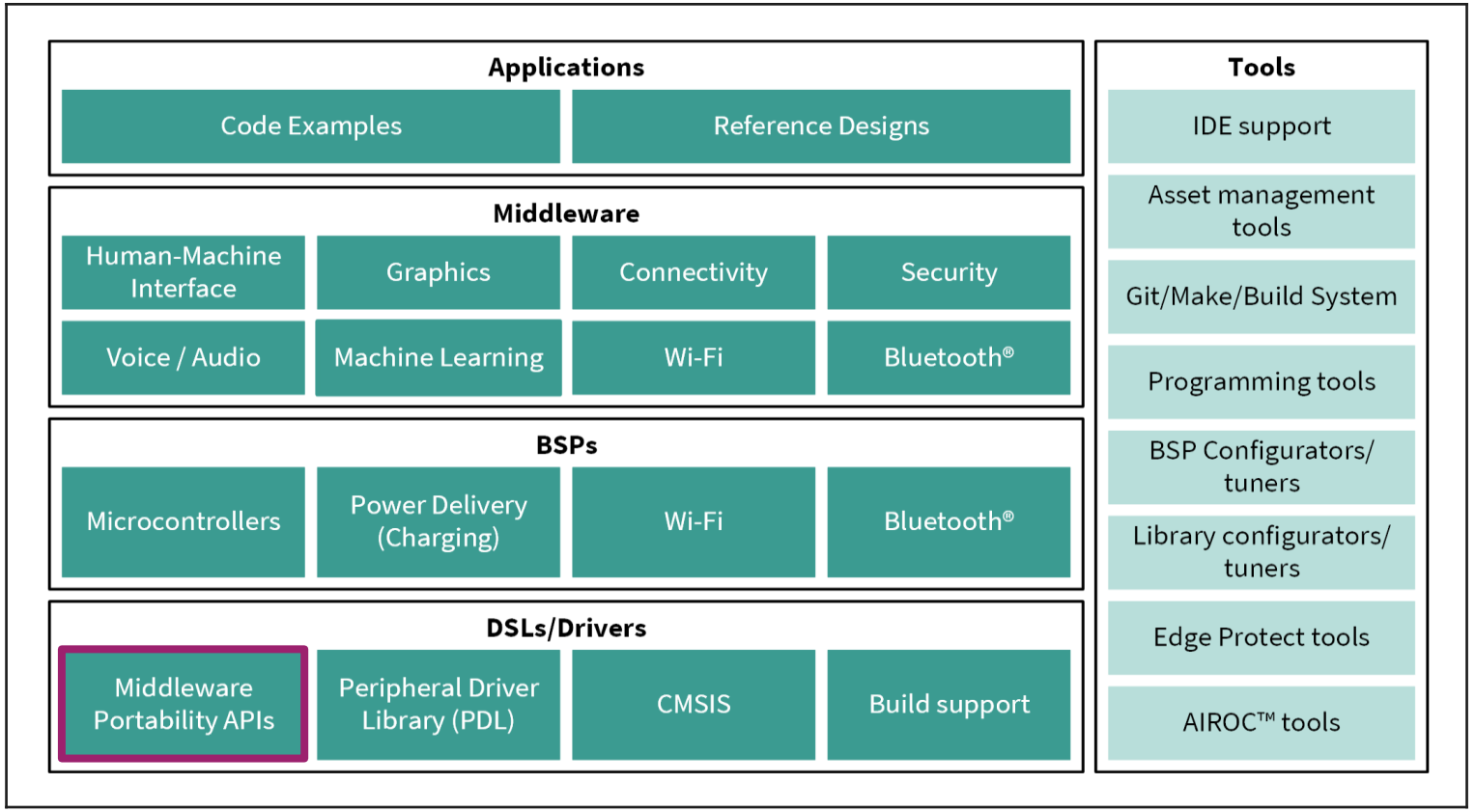


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Firmware / application development: Software resources (2/4)

Middleware portability APIs

- High-level APIs built on top of PDL
- Hardware generic, so more portable
- Architected to enable Infineon maintained middleware to work across a wider range of devices
- Replacement of the former ModusToolbox™ HAL layer

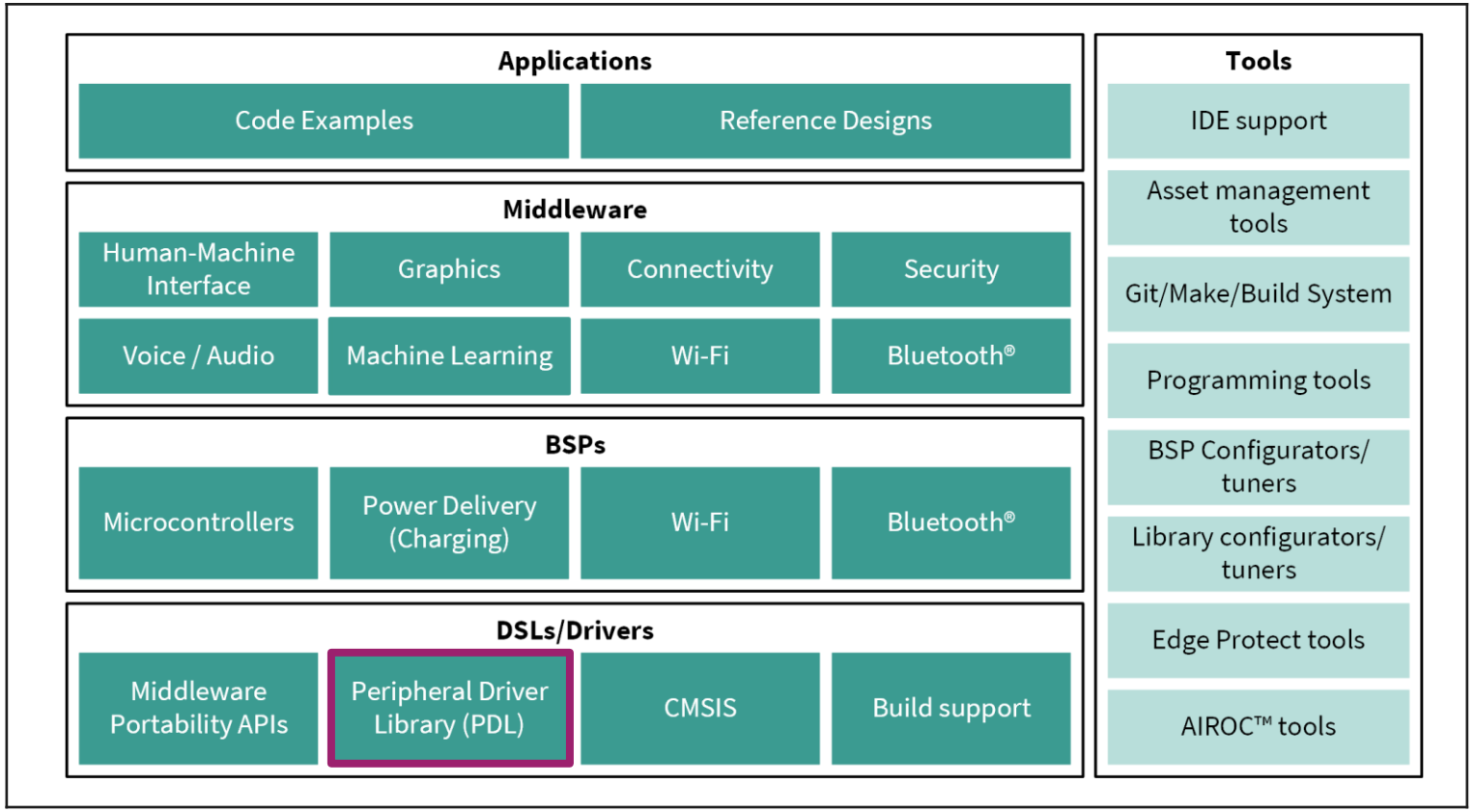


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Firmware / application development : Software resources (3/4)

Peripheral driver library (PDL)

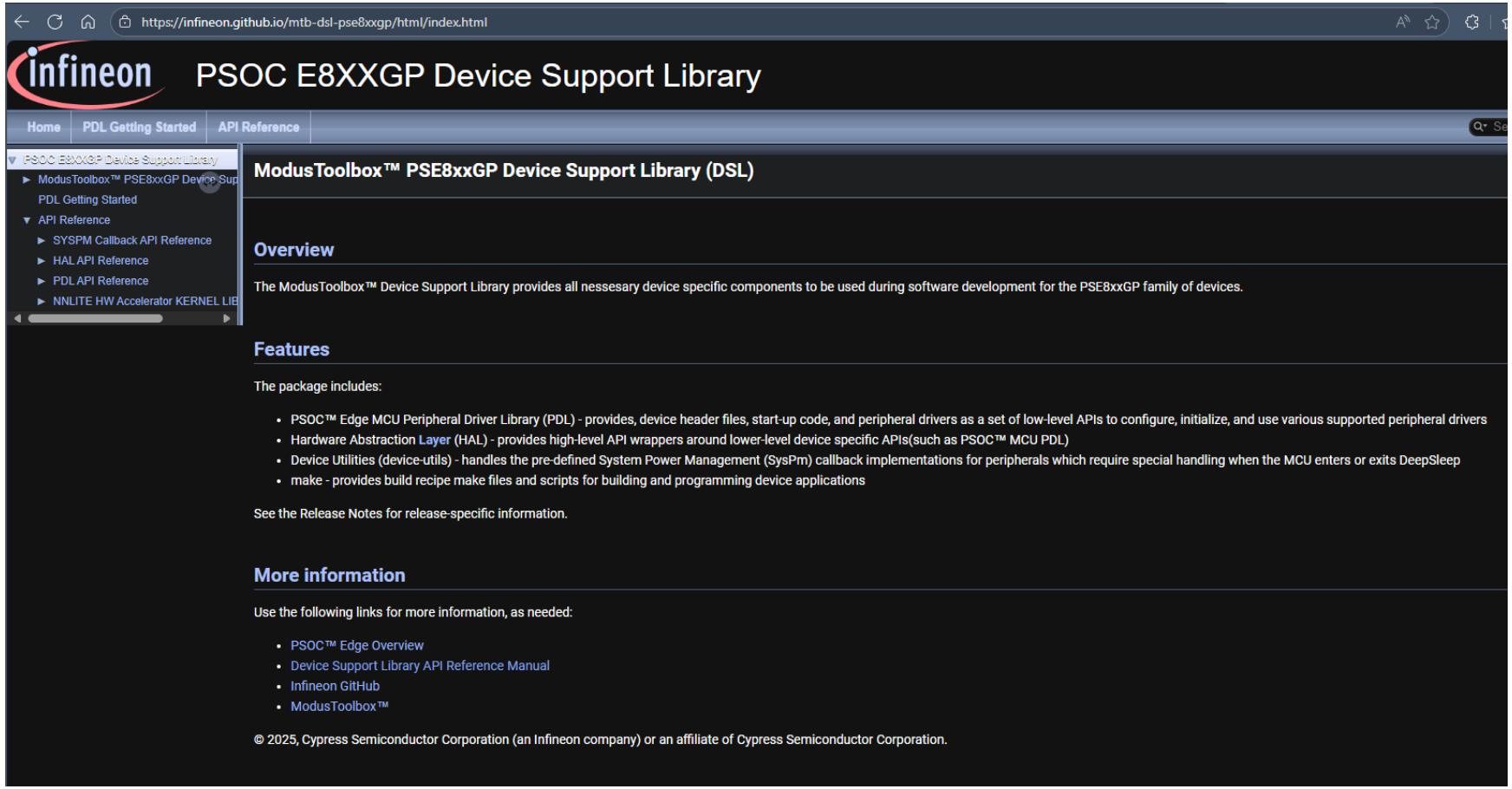
- Set of low-level APIs
- Hardware specific so less portable



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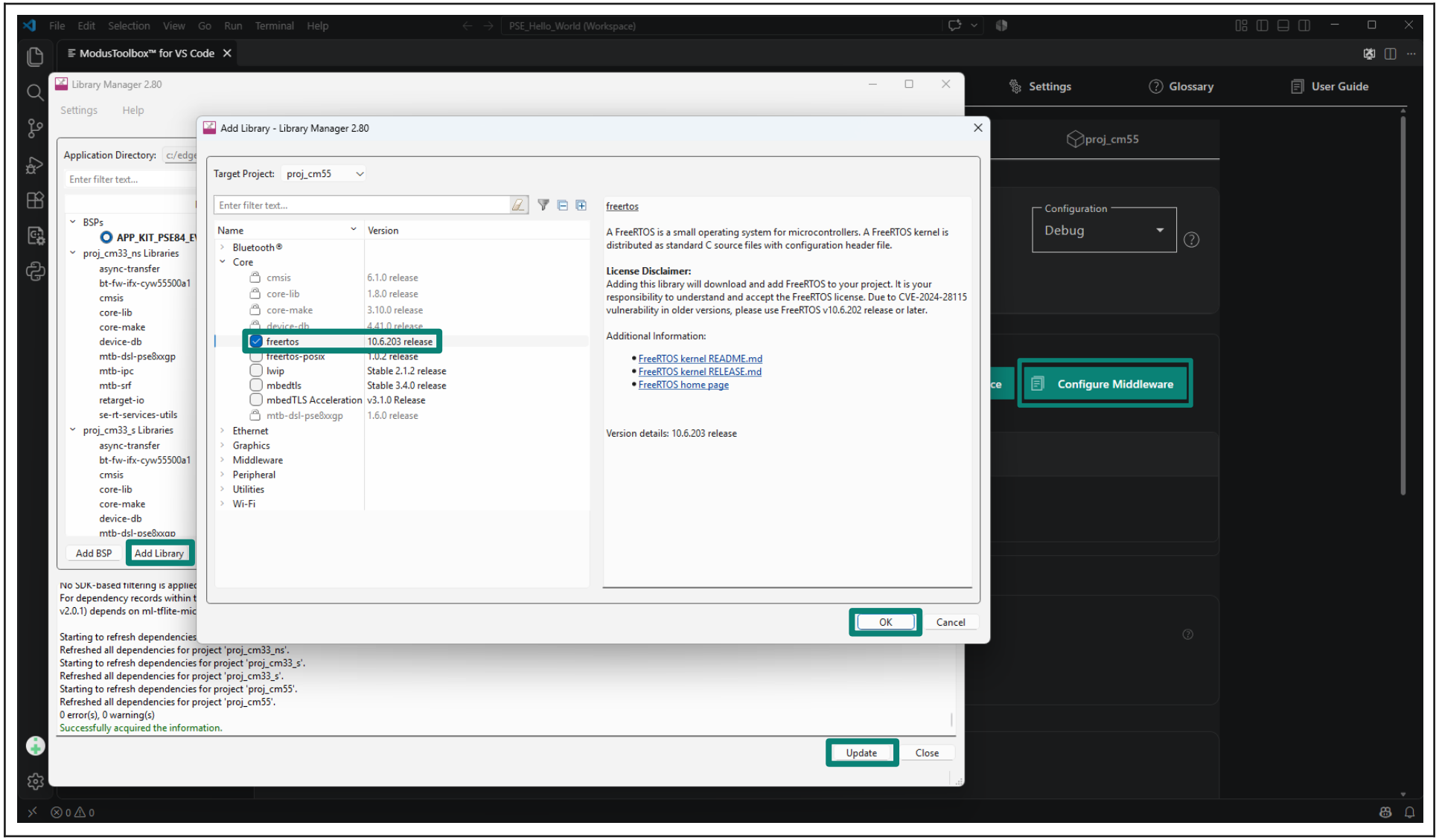
Firmware/application development : Software resources (4/4)

For more information, go to: https://infineon.github.io/mtb-dsl-pse8xxgp/html/index.html



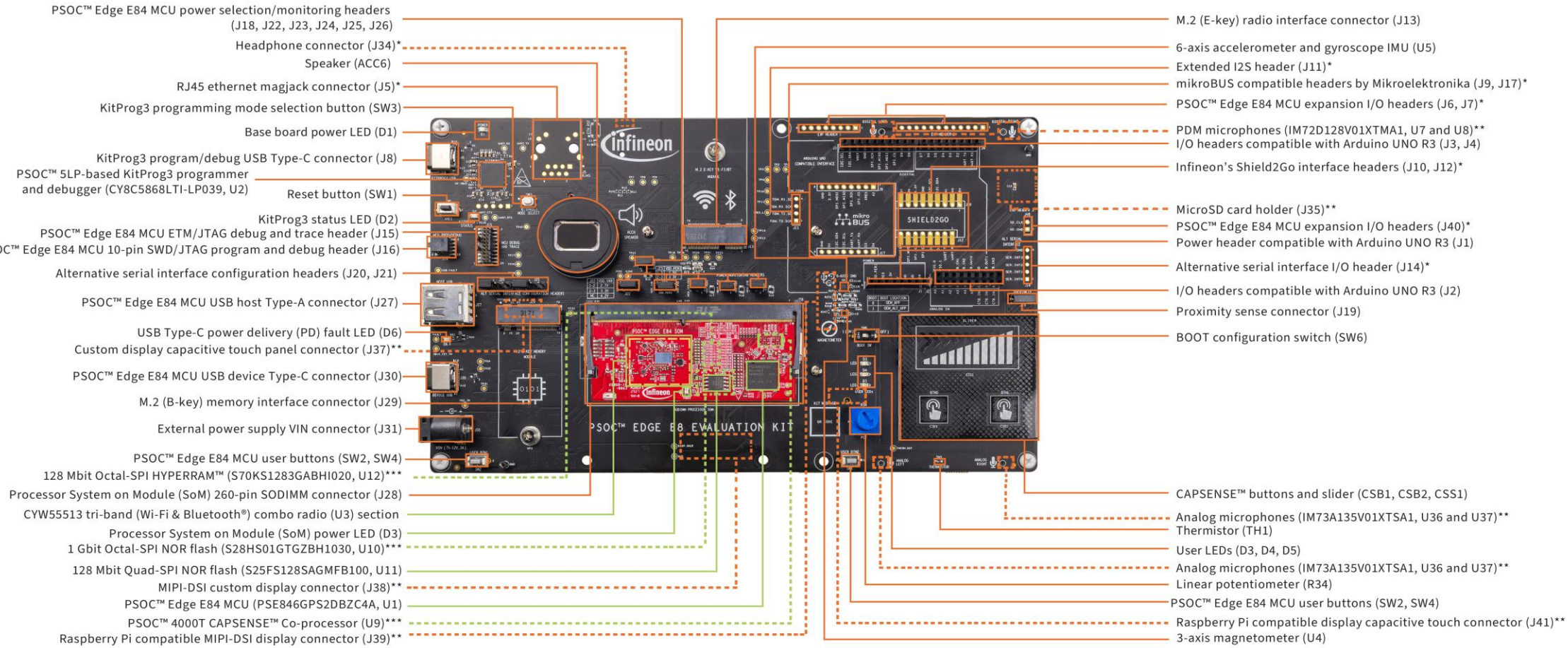
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FreeRTOS support with ModusToolbox™



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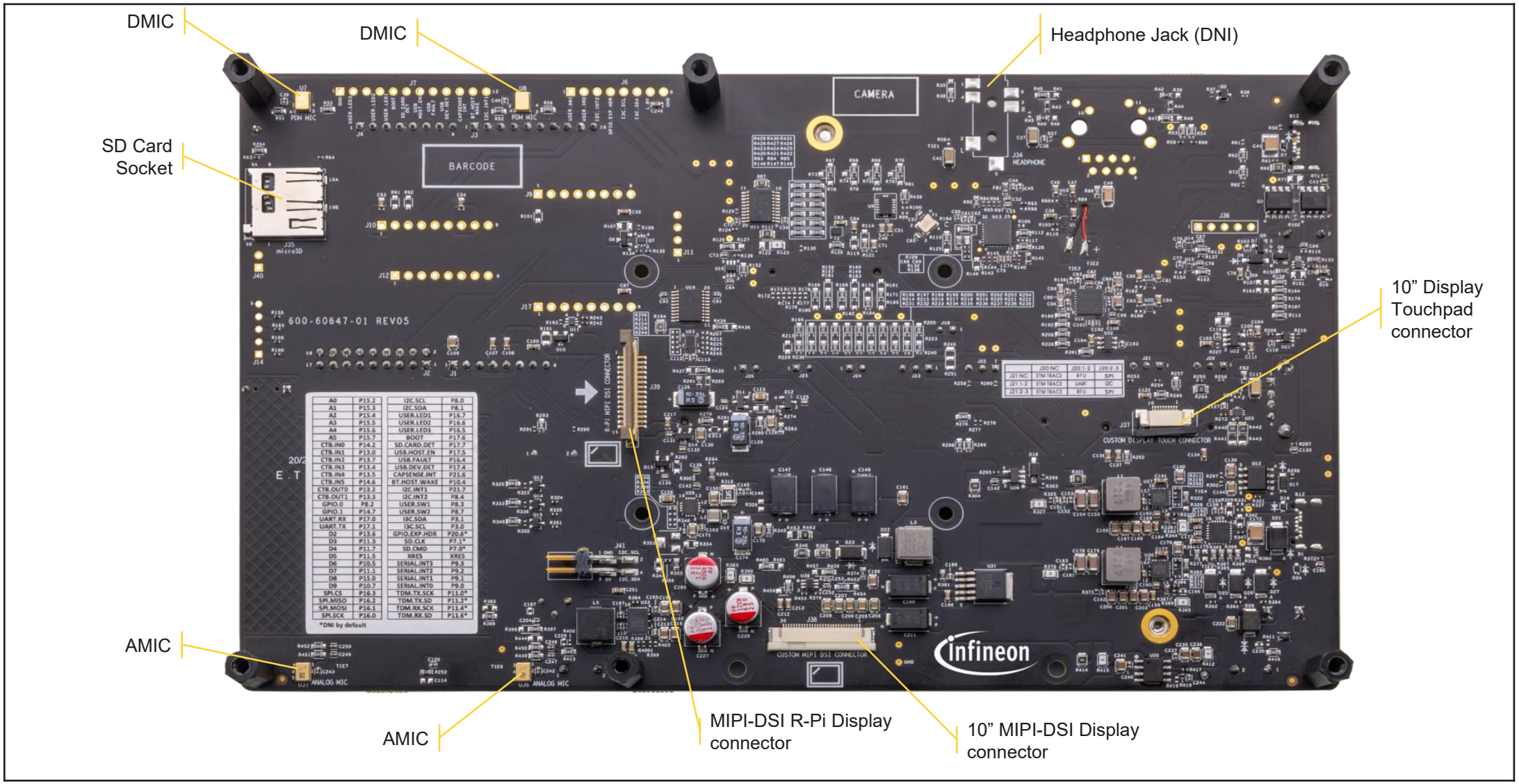
PSOC™ Edge E84 MCU development kit: Top view



*Footprint only, not populated on the board **Component at the bottom side of the Baseboard ***Component at the bottom side of the SoM

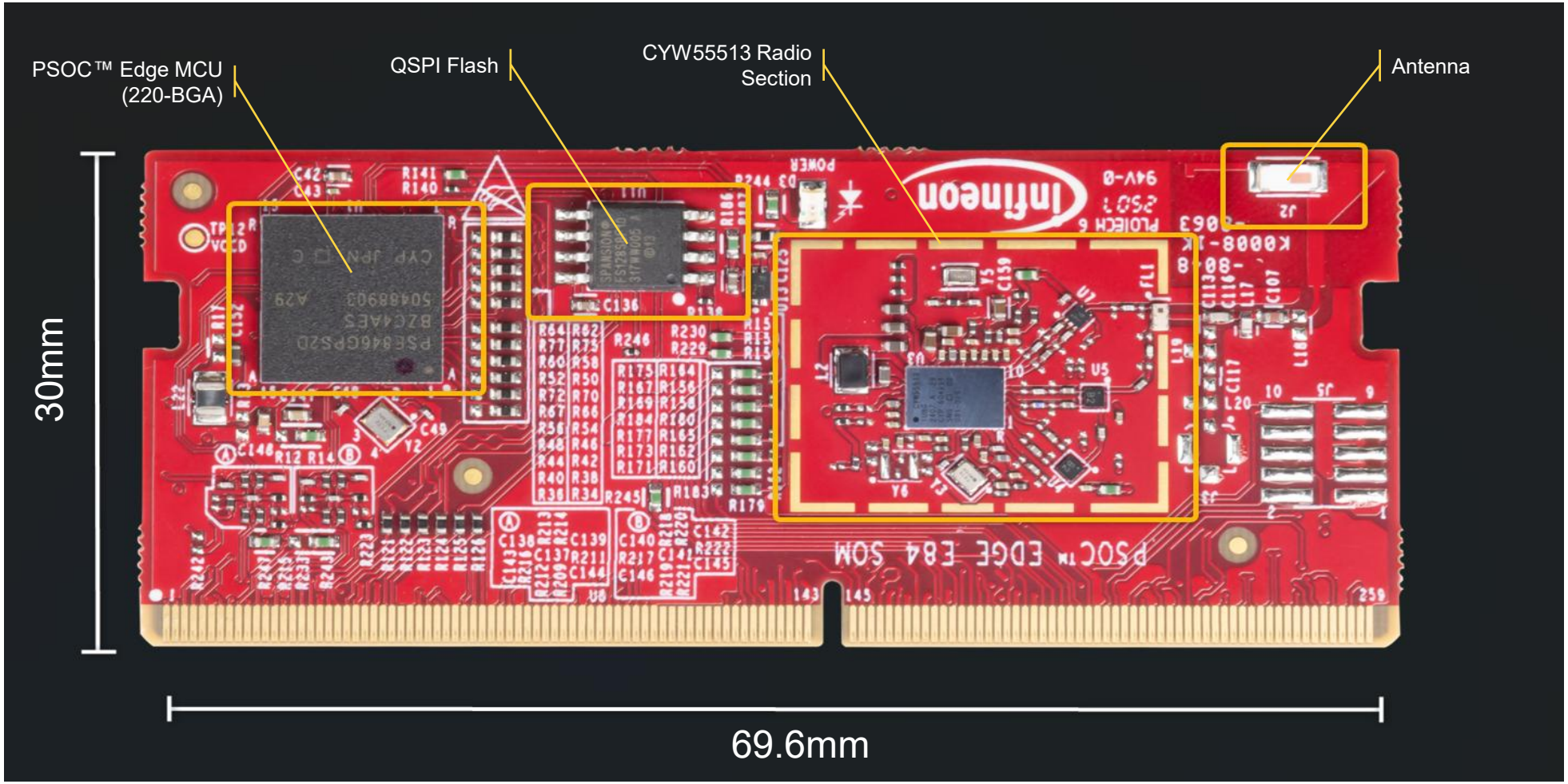
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PSOC™ Edge E84 MCU development kit: Back view



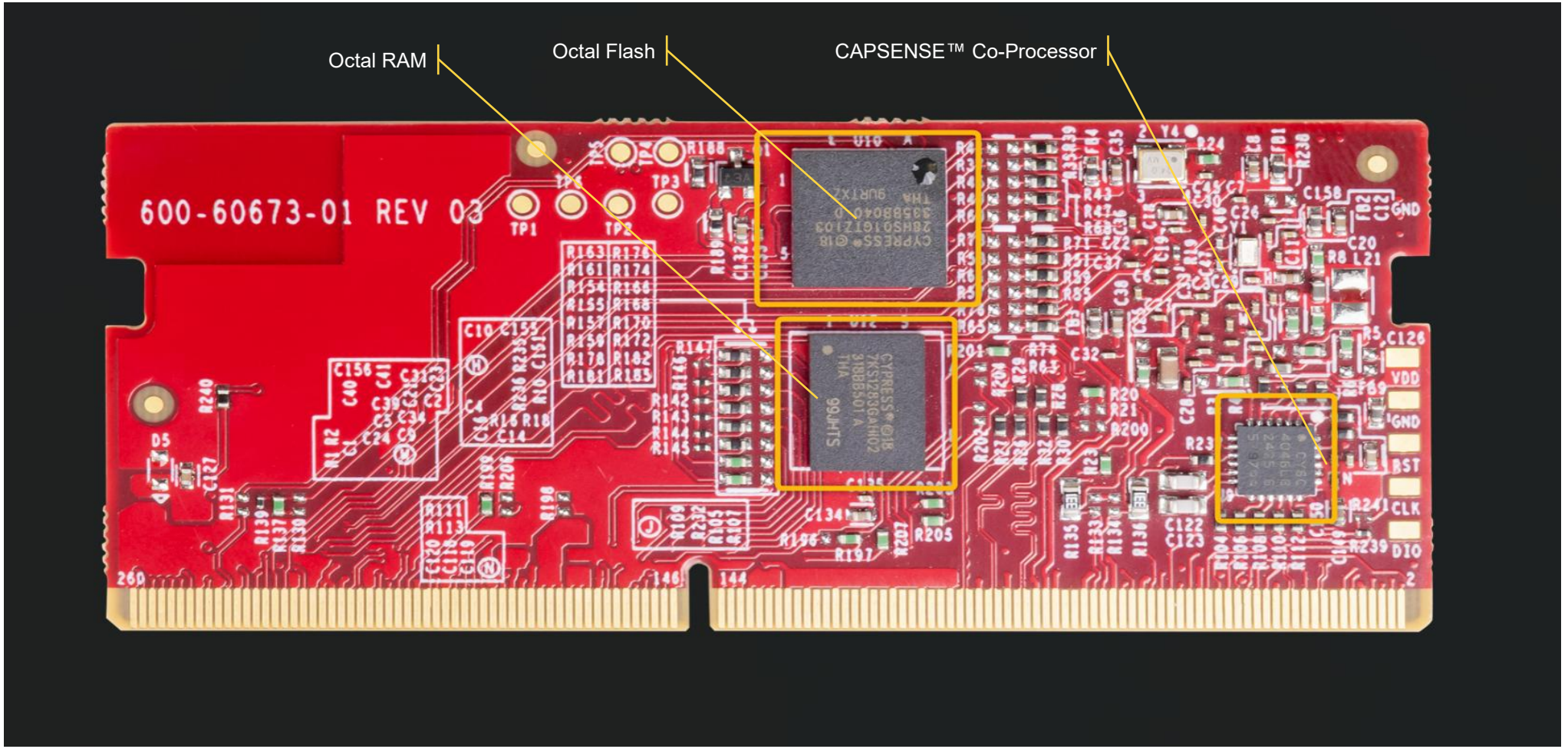
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PSOC™ Edge E84 MCU SOM : Top view



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PSOC™ Edge E84 MCU SOM: Bottom view



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Programming/debugging

- Programming and Debugging PSOC™ Edge E84 MCU
  - Comes with a KitProg3 onboard programmer and debugger
  - Supports Cortex Microcontroller Software Interface Standard – Debug Access Port CMSIS-DAP
  - Supports Open On-Chip Debugger (OpenOCD) using a GDB server and the J-Link debug probe
  - Debug environment is IDE-specific – see the ModusToolbox™ documentation

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2 **Code Example 1: Hello World + Blinky** **46**

3 Code Example 2: IPC Semaphores using PDL API 50

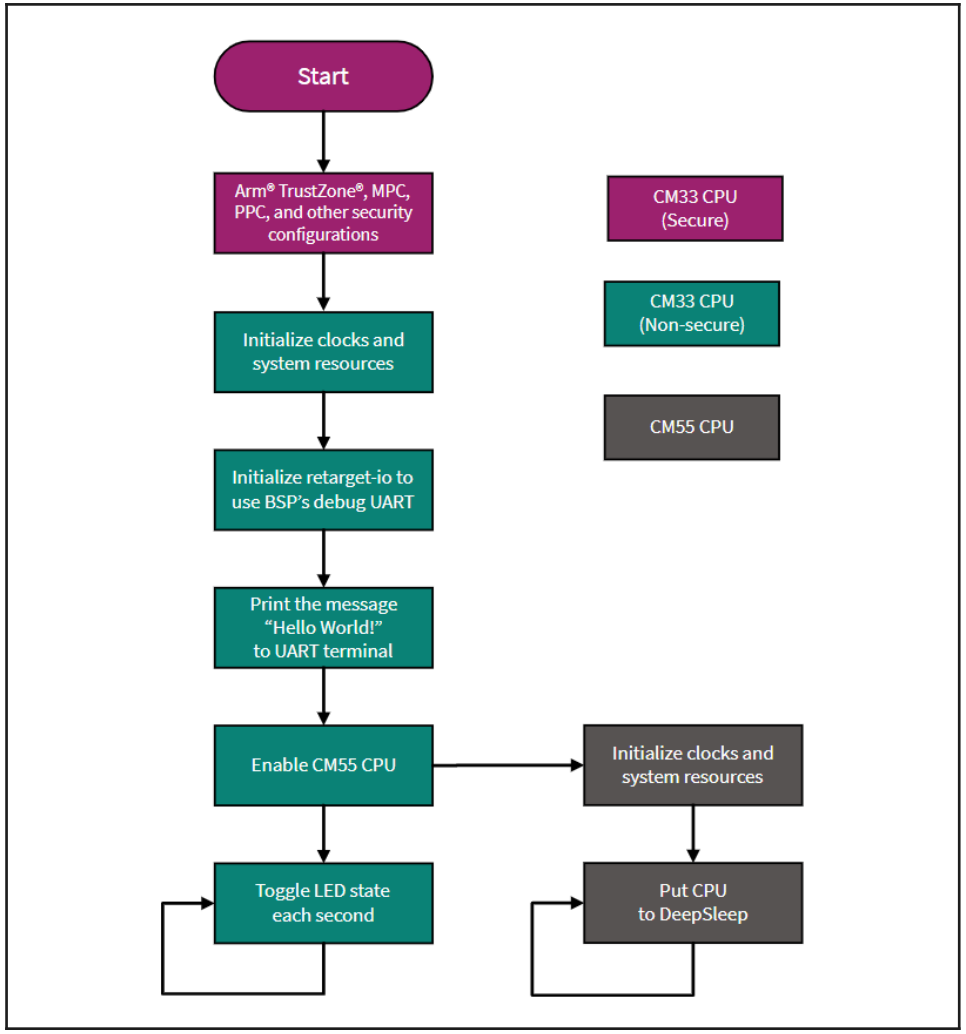
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Lab 1: Hello world + Blinky

What will you learn ?

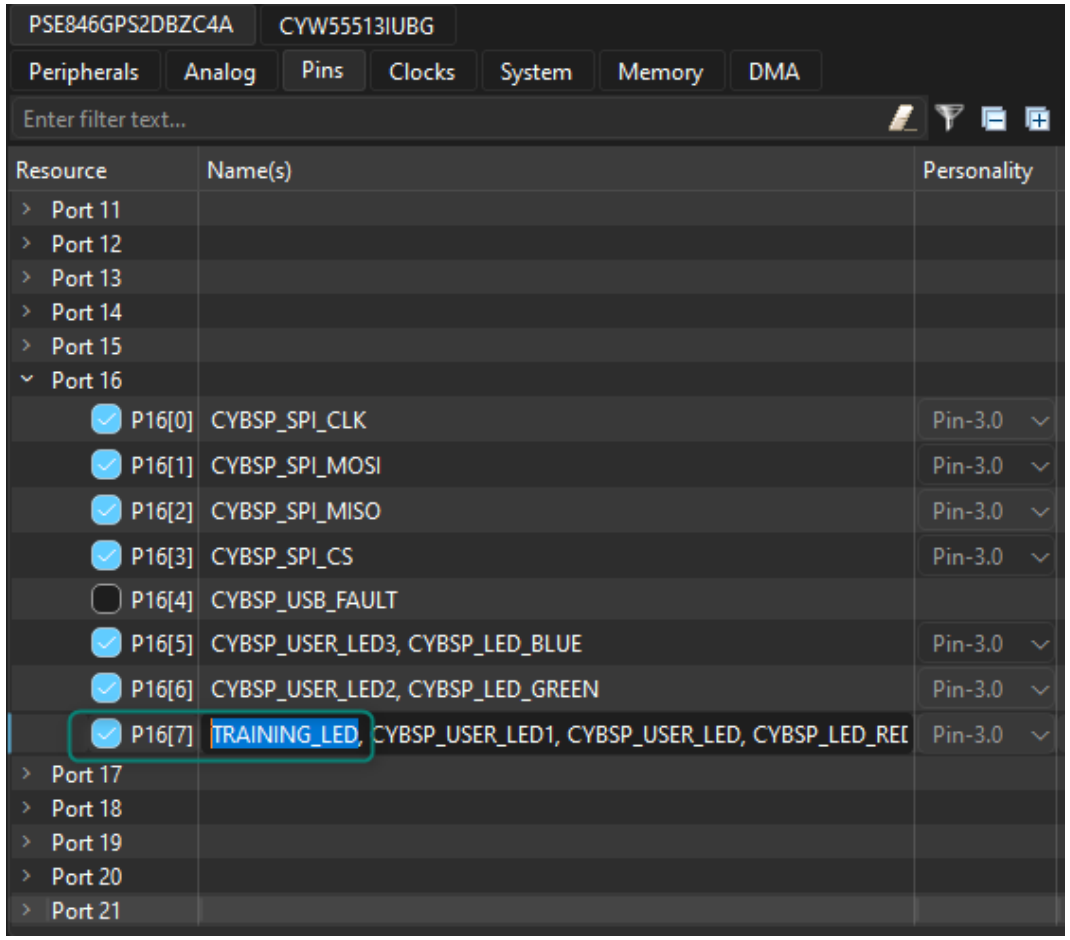
- How to run a basic Hello World example
- Understanding the structure of PSOC™ Edge projects
- How to rename pins using Device Configurator

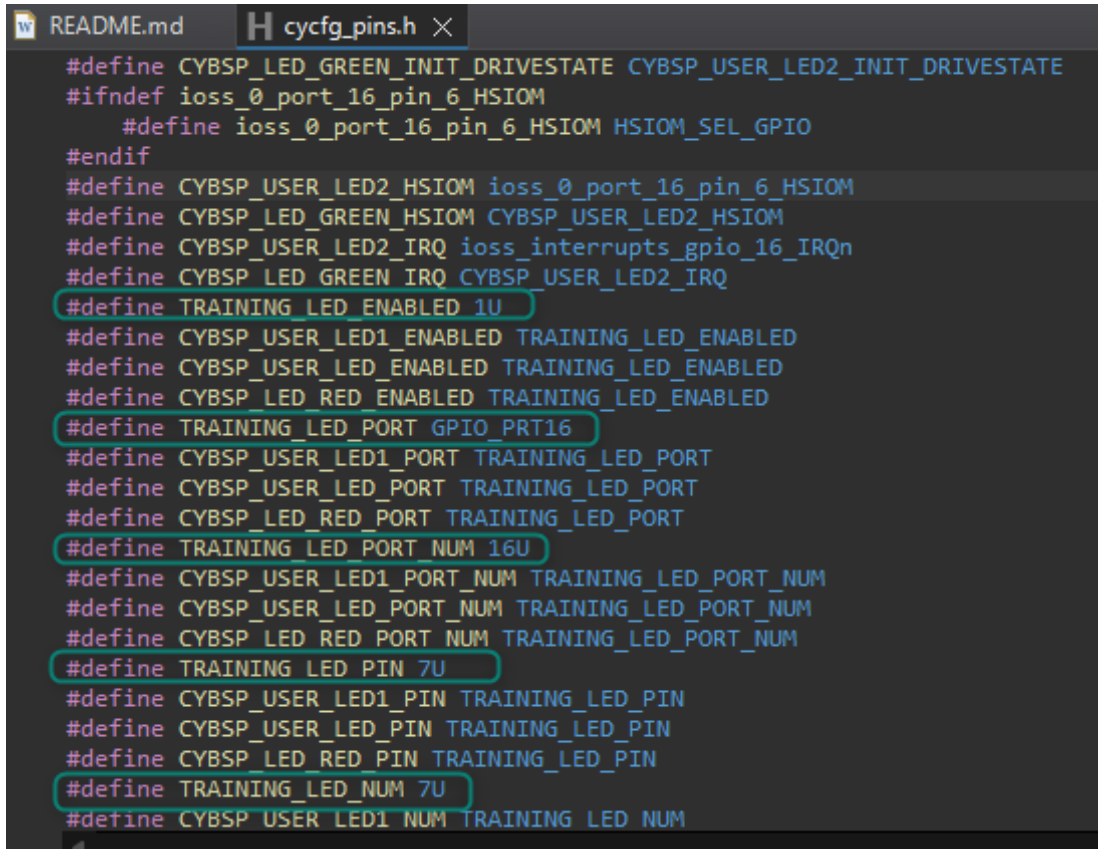
Description of Hello World + Blinky example



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Using Device Configurator





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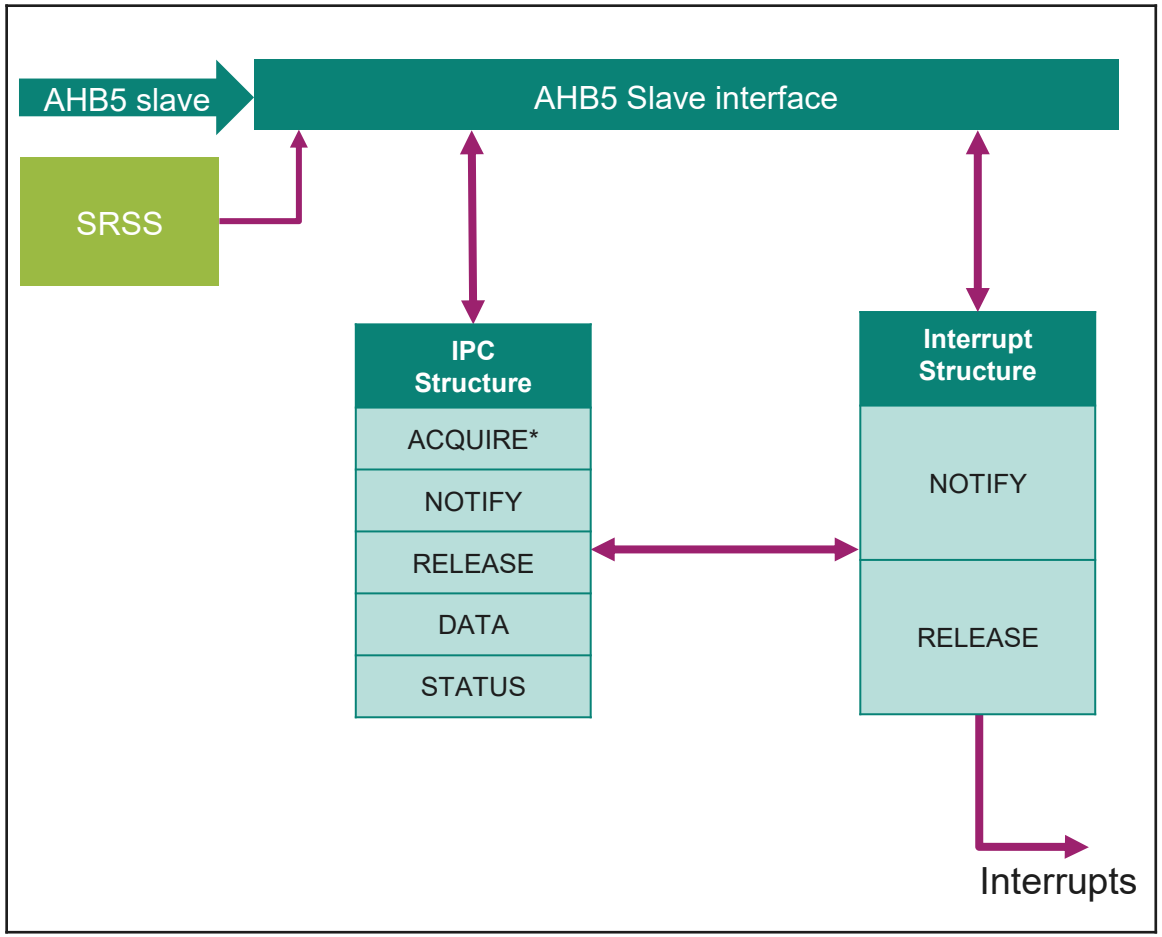
LAB2: IPC Semaphores using PDL

What will you learn ?

- How to use the Library Manager in ModusToolbox™
- How to share a resource between CPUs
- How to use IPC semaphores to synchronize a shared resource

Overview of IPC

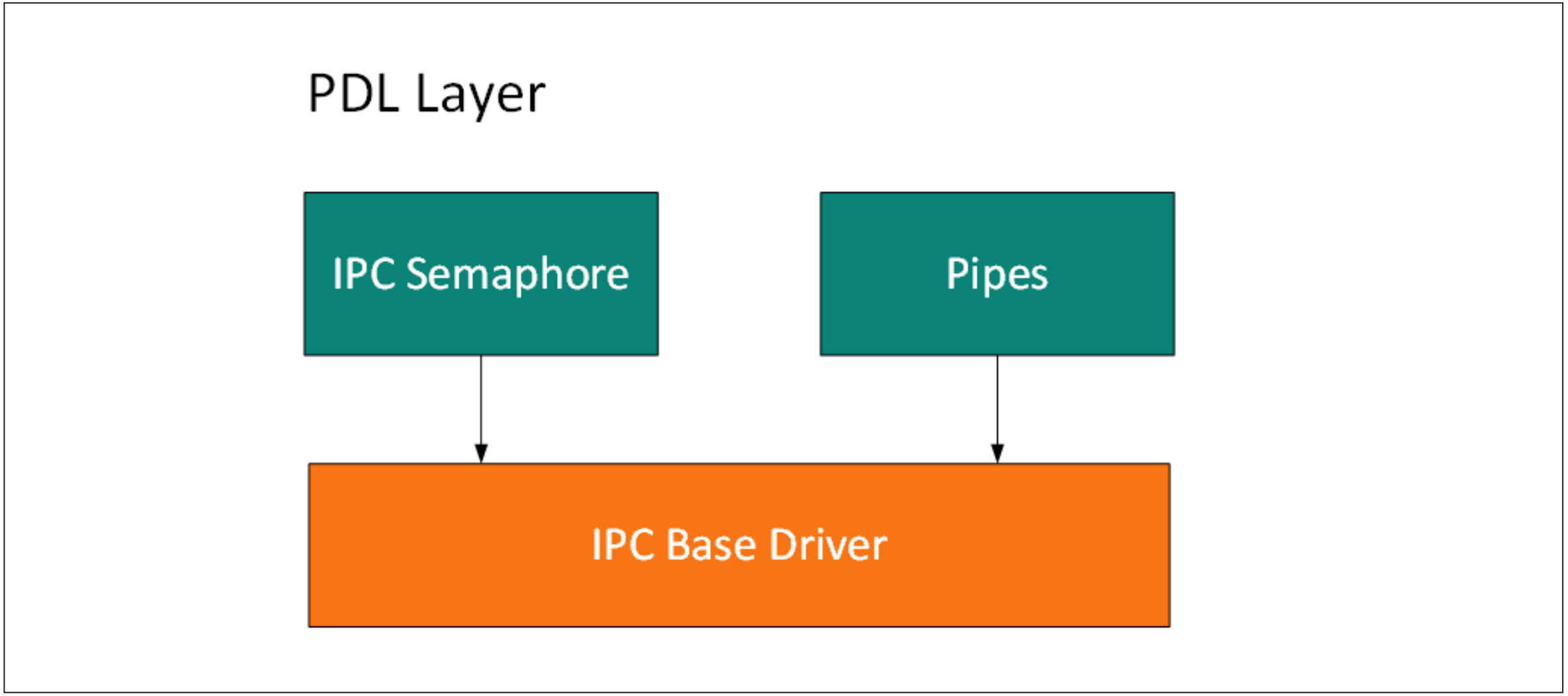
- Supports sixteen communication channels
- Supports sixteen interrupt structures
- Inter-processor communication methods
  - Half-duplex with shared event handling
  - Half-duplex with independent event handling
  - Full-duplex communication
- IPC is available in CPU active power mode only



* MS field of the Acquire register width has been expanded to 8 bits

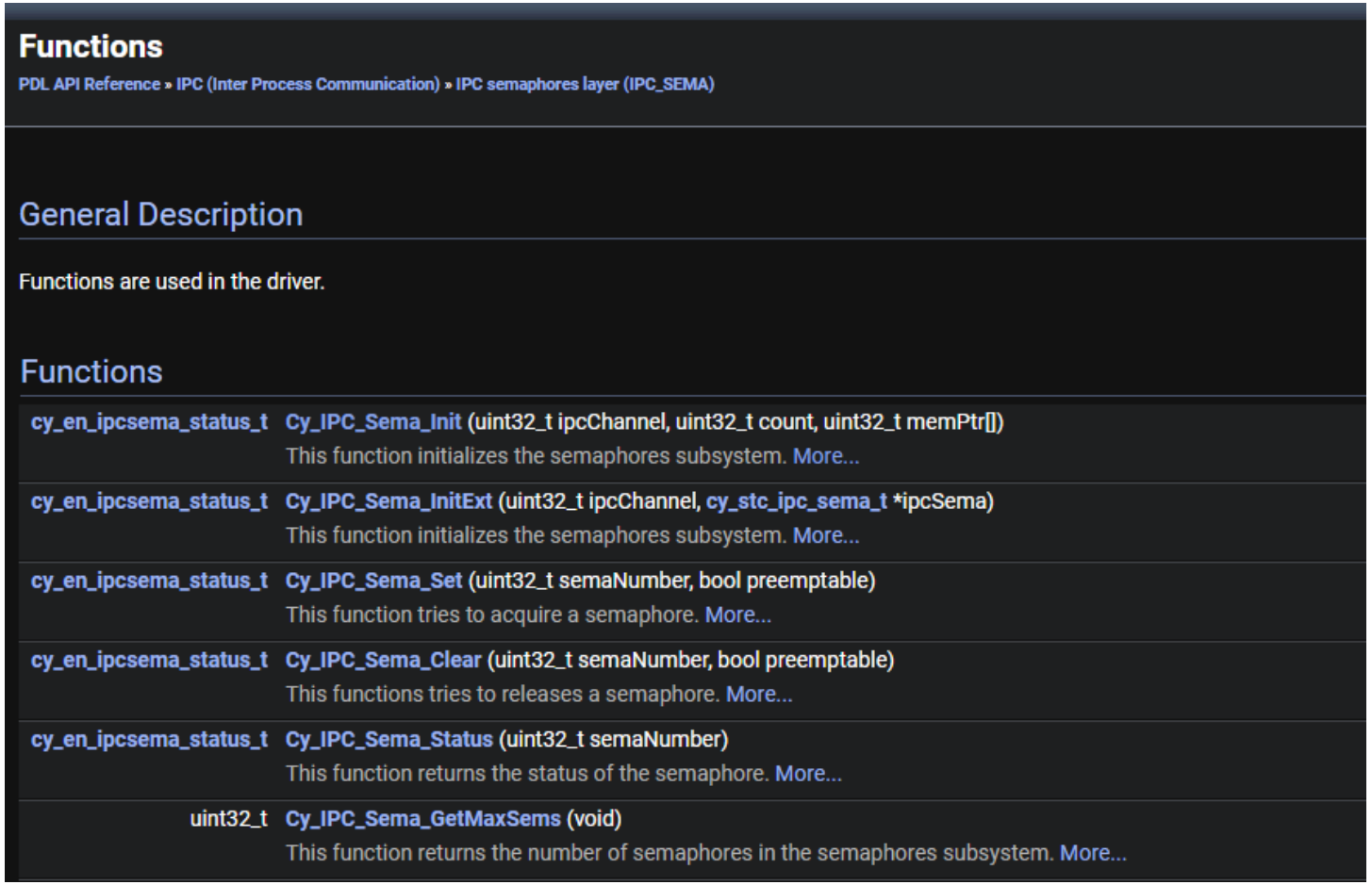
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Block diagram of IPC PDL



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IPC PDL (Peripheral abstraction layer) functions



Functions

PDL API Reference » IPC (Inter Process Communication) » IPC semaphores layer (IPC_SEMA)

General Description

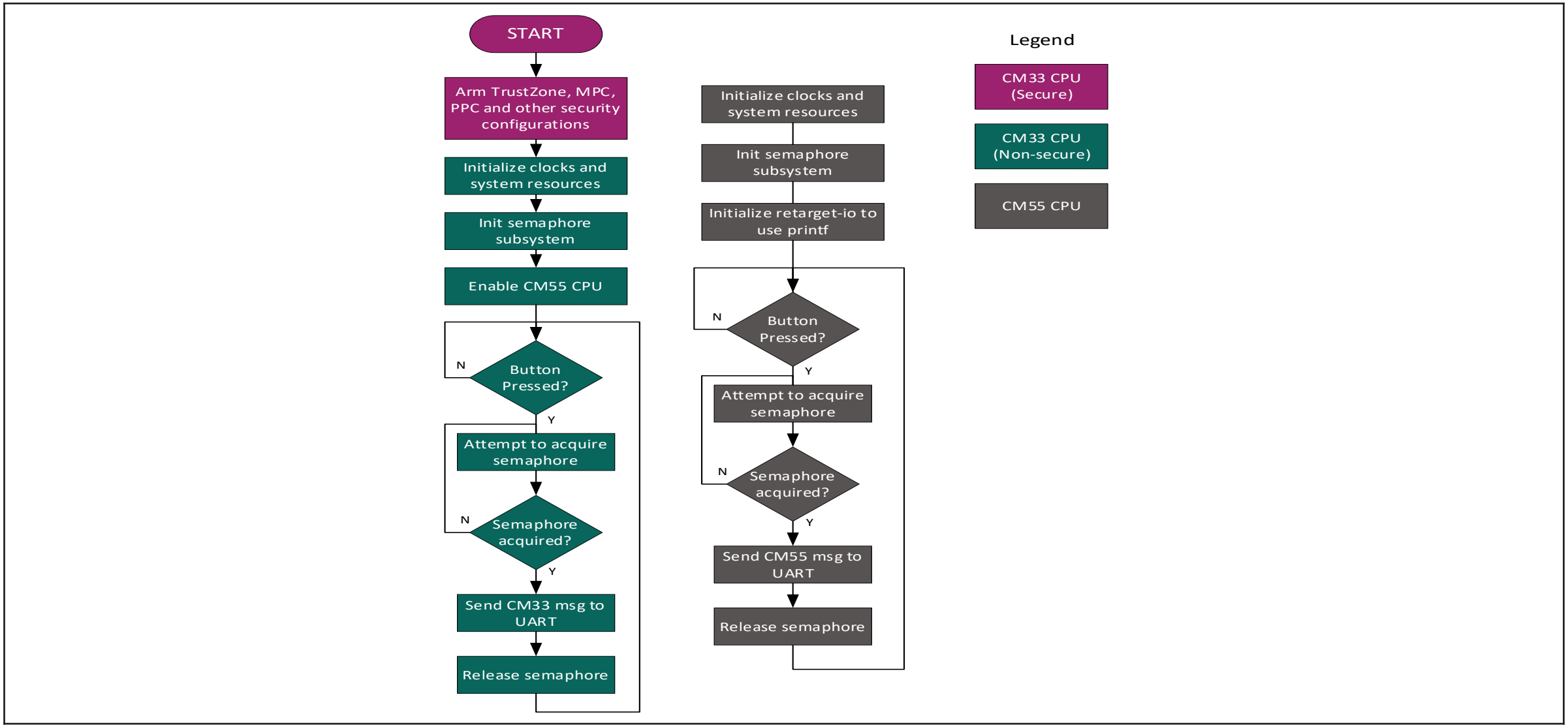
Functions are used in the driver.

Functions

cy_en_ipcsema_status_t	**Cy_IPC_Sema_Init** (uint32_t ipcChannel, uint32_t count, uint32_t memPtr[])
	This function initializes the semaphores subsystem. More...
cy_en_ipcsema_status_t	**Cy_IPC_Sema_InitExt** (uint32_t ipcChannel, cy_stc_ipc_sema_t *ipcSema)
	This function initializes the semaphores subsystem. More...
cy_en_ipcsema_status_t	**Cy_IPC_Sema_Set** (uint32_t semaNumber, bool preemptable)
	This function tries to acquire a semaphore. More...
cy_en_ipcsema_status_t	**Cy_IPC_Sema_Clear** (uint32_t semaNumber, bool preemptable)
	This functions tries to releases a semaphore. More...
cy_en_ipcsema_status_t	**Cy_IPC_Sema_Status** (uint32_t semaNumber)
	This function returns the status of the semaphore. More...
uint32_t	**Cy_IPC_Sema_GetMaxSems** (void)
	This function returns the number of semaphores in the semaphores subsystem. More...

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Flow chart



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Revision history

Document revision	Date	Description of change
**	2025-11-25	Initial release.
*A	2026-03-15	Migrated to the latest empower training presentation template.
*B	2026-09-17	Update tool versions. Use VSCode for hands-on labs.

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